

Postural adaptations may contribute to the unique locomotor energetics seen in hopping kangaroos

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This **valuable** biomechanical analysis of kangaroo kinematics and kinetics across a range of hopping speeds and masses is a step towards understanding a long-standing problem in locomotion biomechanics: the mechanism for how kangaroos, unlike other mammals, can increase hopping speed without a concomitant increase in metabolic cost. The authors **convincingly** demonstrate that changes in kangaroo posture with speed increase tendon stress/strain and hence elastic energy storage/return. This greater tendon elastic energy storage/return may counteract the increased cost of generating muscular force at faster speeds and thus allows for the invariance in metabolic cost. This methodologically impressive study sets the stage for further work to investigate the relation of hopping speed to metabolic cost more definitively.

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Abstract

Hopping kangaroos exhibit remarkably little change in their rate of metabolic energy expenditure with locomotor speed compared to other running animals. This phenomenon may be related to greater elastic energy savings due to increasing tendon stress; however, the mechanisms which enable the rise in stress without additional muscle work remain poorly understood. In this study, we created a three-dimensional (3D) kangaroo musculoskeletal model, integrating 3D motion capture and force plate data, to analyse the kinematics and kinetics of hopping red and grey kangaroos. Using our model, we evaluated how body mass and speed influence (i) hindlimb posture, (ii) effective mechanical advantage (EMA), and (iii) the associated tendon stress in the ankle extensors and (iv) ankle work during hopping. We found that increasing ankle dorsiflexion and metatarsophalangeal plantarflexion likely

played an important role in decreasing ankle EMA by altering both the muscle and external moment arms, which subsequently increased energy absorption and peak tendon stress at the ankle. Surprisingly, kangaroo hindlimb posture appeared to contribute to increased tendon stress, allowing more energy to be stored at faster speeds. These posture-mediated increases in elastic energy storage and return could be a key factor enabling kangaroos to achieve energetic benefits at faster hopping speeds, but may limit the performance of large kangaroos due to the risk of tendon rupture.

Introduction

Kangaroos and other macropods are unique in both their morphology and their locomotor style. At slow speeds they use a pentapedal gait, where the forelimbs, the hindlimbs, and the tail all contact the ground, while at faster movement speeds they use their distinctive hopping gait (Dawson and Taylor 1973 [DOI](#), O'Connor *et al.* 2014 [DOI](#)). Their uniqueness extends into their energetics of locomotion. As far back as the 19th century, researchers noticed that the metabolic cost of running in quadrupeds and bipeds, like dogs, horses and humans, increased linearly with speed (Zuntz 1897 [DOI](#), Taylor *et al.* 1970 [DOI](#), Heglund *et al.* 1982 [DOI](#), Taylor *et al.* 1982 [DOI](#)). To explain why metabolic rate increased at faster running speeds among quadrupeds and bipeds, Kram and Taylor (1990) [DOI](#) refined the ‘cost of generating force’ hypothesis (Taylor *et al.* 1980 [DOI](#)). They reasoned that the decrease in contact time with increased speed, reflects an increase in the rate of generating muscle force, and the rate of cross-bridge cycling (Kram and Taylor 1990 [DOI](#)). This was supported for a diverse range of running and hopping animals, suggesting that metabolic rate was inversely proportional to contact time (Kram and Taylor 1990 [DOI](#)). Yet hopping macropods appear to defy this trend. On treadmills, both red kangaroos (~20 kg) and tammar wallabies (~5 kg) showed little to no increase in the rate of oxygen consumption with increased hopping speed (Dawson and Taylor 1973 [DOI](#), Baudinette *et al.* 1992 [DOI](#), Kram and Dawson 1998 [DOI](#)). The underlying mechanisms explaining how macropods are able to uncouple hopping speed and energy cost is not completely understood (Thornton *et al.* 2022 [DOI](#)).

The ability to uncouple speed and energy expenditure in macropods is likely related to the behaviour of their ankle extensor muscle-tendon units, which store and return elastic strain energy (Morgan *et al.* 1978 [DOI](#), Biewener *et al.* 2004b [DOI](#), McGowan *et al.* 2005 [DOI](#)). In tammar wallabies, ankle tendon stress increased with hopping speed, leading to a greater rise in elastic strain energy return than muscle work, which increases the proportion of work done by tendon recoil while muscle work remains near constant (Baudinette and Biewener 1998 [DOI](#), Biewener *et al.* 1998 [DOI](#)). Size-related differences in ankle extensor tendon morphology (Bennett and Taylor 1995 [DOI](#), McGowan *et al.* 2008 [DOI](#)), and the resultant low strain energy return, may explain why small (<3 kg) hopping macropods and rodents appear not to be afforded the energetic benefits observed in larger macropods (Thompson *et al.* 1980 [DOI](#), Biewener *et al.* 1981 [DOI](#), Biewener *et al.* 1998 [DOI](#)) (but see Christensen *et al.* (2022) [DOI](#)). However, tendon morphology alone is insufficient to explain why large macropods can increase speed without cost, while large quadrupeds with similar tendon morphology cannot (Dawson and Webster 2010 [DOI](#)). The most obvious difference between macropods and other mammals is their hopping gait, but previously proposed mechanisms to explain how hopping could reduce the metabolic cost of generating muscle force, such as near-constant stride frequency (Heglund and Taylor 1988 [DOI](#), Dawson and Webster 2010 [DOI](#)) or respiratory-stride coupling (Baudinette *et al.* 1987 [DOI](#)), do not distinguish between small and large macropods, nor galloping quadrupeds (McGowan and Collins 2018 [DOI](#)).

Postural changes are another mechanism which could contribute to reduced energetic costs by altering the leverage of limb muscles. The effective mechanical advantage (EMA) is the ratio of the internal muscle-tendon moment arm (the perpendicular distance between the muscle’s force-generating line-of-action and the joint centre) and the ground reaction force (GRF) moment arm (the perpendicular distance between the vector of the GRF to the joint centre). Smaller EMA

requires greater muscle force to produce a given force on the ground, thereby demanding a greater volume of active muscle, and presumably greater metabolic rates, than larger EMA for the same physiology. In humans, an increase in limb flexion and decrease in limb EMA recruits a greater volume of active muscle and increases the metabolic cost of running compared to walking (Biewener *et al.* 2004a [↗](#), Kipp *et al.* 2018 [↗](#), Allen *et al.* 2022 [↗](#)). Changes in EMA with speed are not common in quadrupedal mammals (Biewener 1989 [↗](#)) but these are not without precedent. Elephants—one of the largest and noticeably most upright animals—do not have discrete gait transitions, but rather continuously and substantially decrease EMA in limb joints when moving faster, which has also been linked to the increase in metabolic cost of locomotion with speed (Ren *et al.* 2010 [↗](#), Langman *et al.* 2012 [↗](#)). EMA is more commonly observed to change with size in terrestrial mammals. Smaller animals tend to move in crouched postures with limbs becoming progressively more extended as body mass increases (Biewener 1989 [↗](#)). Yet, rather than metabolic cost, this postural transition appeared to be driven by the need to reduce the size-related increases in tissue stress (Dick and Clemente 2017 [↗](#), Clemente and Dick 2023 [↗](#)). As a consequence of transitioning to more upright limb postures, musculoskeletal stresses in terrestrial mammals are independent of body mass (Biewener 1989 [↗](#), 2005).

Macropods, in contrast, maintain a crouched posture during hopping. Kram and Dawson (1998 [↗](#)) explored whether kangaroos transitioned to a more extended limb posture with increasing speed as a potential mechanism for their constant metabolic rate, but did not detect a change in EMA at the ankle across speeds from 4.3 to 9.7 m s⁻¹. Further, ankle posture and EMA appears to vary weakly (McGowan *et al.* 2008 [↗](#)) or not at all with body mass in macropods (Bennett and Taylor 1995 [↗](#), Snelling *et al.* 2017 [↗](#)). As a consequence, stress would be expected to increase with both hopping speed and body mass. Large tendon stresses are required for rapid strain energy return (strain energy $\propto$ stress²) (Biewener and Baudinette 1995 [↗](#)). Given that tendon recoil plays a pivotal role in hopping gaits, it is perhaps unsurprising that tendon stress approaches the safe limit in larger kangaroos. The ankle extensor tendons in a moderately sized male red kangaroo (46.1 kg) operate with safety factors near two even at slow hopping speeds (3.9 m s⁻¹) (Kram and Dawson 1998 [↗](#)), far lower than the typical safety factor of four to eight for mammalian tendons (Ker *et al.* 1988 [↗](#)). Tendon stresses are also unusually large in smaller kangaroos. Juvenile and adult western grey kangaroos, ranging in mass from 5.8 to 70.5 kg, all hop with gastrocnemius and plantaris tendon safety factors less than two (Snelling *et al.* 2017 [↗](#)). Large tendon stresses may not only be a natural consequence of their crouched posture and tendon morphology, but also adaptively selected. Considering this, kangaroos may adjust their posture to increase tendon stress and its associated elastic energy return. If so, there would likely be systematic variation in kangaroo posture with speed and mass which is yet to be fully explored.

In this study, we investigate the hindlimb kinematics and kinetics in kangaroos hopping at various speeds. Specifically, we explore the relationship between changes in posture, EMA, joint work, and tendon stress across a range of hopping speeds and body masses. To do this, we built a musculoskeletal model of a kangaroo based on empirical imaging and dissection data (Fig. 1a [↗](#)). We used the musculoskeletal model to calculate ankle EMA throughout the stride by capturing changes in both muscle moment arm and the GRF moment arm, allowing us to explore their individual contributions. We hypothesised that (i) the hindlimb would be more crouched at faster speeds, primarily due to the distal hindlimb joints (ankle and metatarsophalangeal), independent of changes with body mass, and (ii) changes in moment arms resulting from the change in posture would contribute to the increase in tendon stress with speed, and may thereby contribute to energetic savings by increasing the amount of positive and negative work done by the ankle without requiring additional muscle work.

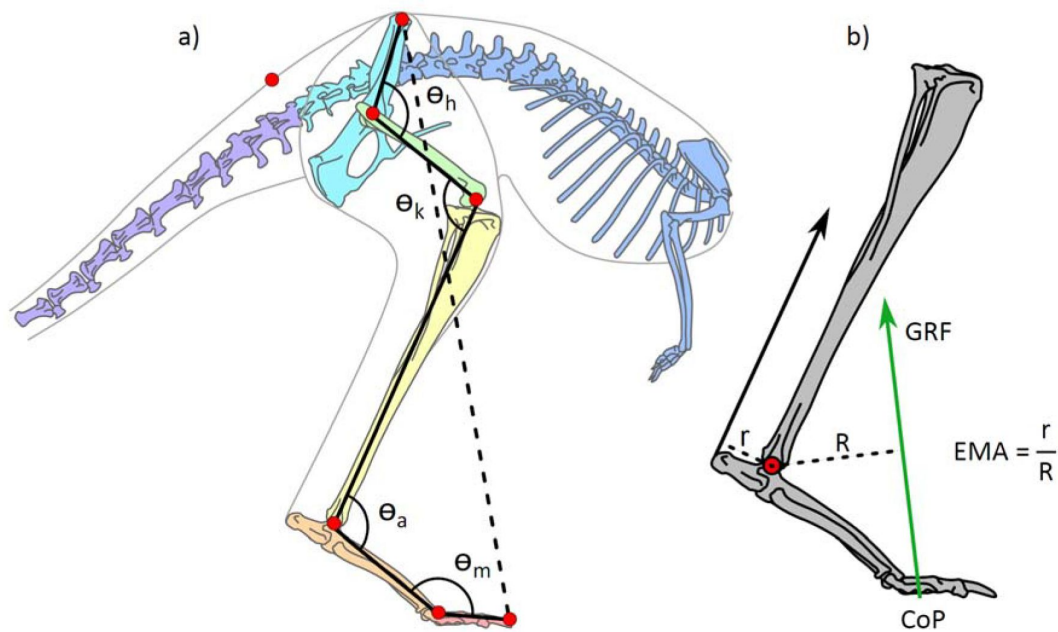


Figure 1

(a) Illustration of the kangaroo model. Total leg length was calculated as the sum of the segment lengths (solid black lines) in the hindlimb and compared to the pelvis-to-toe distance (dashed line) to calculate the crouch factor. Joint angles were determined for the hip, h, knee, k, ankle, a, and metatarsophalangeal, m, joints. The model markers (red circles) indicate the position of the reflective markers placed on the kangaroos in the experimental trials and were used to characterise the movement of segments in the musculoskeletal model. (b) Illustration of ankle effective mechanical advantage, EMA, muscle moment arm, r , and external moment arm, R , as the perpendicular distance to the Achilles tendon line of action and ground reaction force (GRF) vector, respectively. The centre of pressure (CoP) was allowed to move in the fore-aft direction.

Results

Ground reaction forces

The stance begins with a braking phase (negative fore-aft horizontal component), followed by a propulsive phase (positive fore-aft horizontal component). At the transition from negative to positive, naturally the GRF was vertical, and this occurred at $42.9 \pm 26.9\%$ of stance. The peak in vertical GRF, however, occurred at $46.4 \pm 5.3\%$ of stance (**Fig. 2a,b**).

Larger kangaroos and faster speeds were associated with greater magnitudes of peak GRFs (**Fig. 2a,b**; **Suppl. Table 2**). When peak vertical GRF was normalised to body weight, peak forces ranged from 1.4 multiples of body weight (BW) to 3.7 BW per leg (mean: 2.41 ± 0.396 BW) and increased with speed (**Suppl. Fig. 2a**). Smaller kangaroos may experience disproportionately greater GRFs at faster speeds (**Suppl. Fig. 2b**, **Suppl. Table 2**). The increase in GRF will tend to increase tendon stress, irrespective of changes in posture.

Kinematics and posture

Kangaroo hindlimb stance phase kinematics varied with both body mass and speed. In partial support of hypothesis (i), greater masses and faster speeds were associated with more crouched hindlimb postures (**Fig. 3a,c**; where crouch factor is the ratio of total limb length to pelvis to toe distance). Size-related changes in posture occurred throughout stance and were distributed as small changes among all hindlimb joints rather than a large shift in any one joint (**Fig. 3d,f**; **Suppl. Table 3**). The hip, knee and ankle tended toward more flexion, and the metatarsophalangeal (MTP) toward greater extension (plantarflexion) in larger kangaroos.

In addition to, and independent of, the change in posture with mass, there were substantial postural changes due to speed (**Suppl. Table 3**). Unlike with mass, speed-related changes occurred predominantly during the braking phase, and the changes were concentrated in the ankle and MTP joints, with little to no change in the proximal joints (**Fig. 3c,e,g**; **Suppl. Table 3**). There was a significant decrease in ankle plantarflexion at initial ground contact and increase in dorsiflexion at midstance at faster speeds (**Fig. 3g**; **Suppl. Table 3**). Maximum ankle dorsiflexion occurred at $44.8 \pm 4.5\%$ of stance, and tended to occur $3.9 \pm 0.7\%$ earlier in stance with each 1 m s^{-1} increase in speed ($P < 0.001$). MTP range of motion increased with speed due to an increase in MTP plantarflexion prior to midstance rather than a change in dorsiflexion (**Fig. 3g**).

Effective Mechanical Advantage

Effective mechanical advantage (EMA) of the ankle decreased as the limb became more crouched. The change in EMA with mass and speed was substantial, particularly in the braking period (**Fig. 2c,d**). We evaluated the change in EMA at midstance, as it was approximately the point in the stride where GRF (and therefore tendon stress) was greatest. EMA at midstance decreased with body mass, although we did not detect an effect of speed (**Fig. 2c,d**; **Suppl. Table 5**), which may be due to an insignificant or undetected effect of speed on the external moment arm to the ankle, R . However, examining whether the decrease in EMA was caused by a decrease in the gastrocnemius and plantaris muscle moment arm, r , or an increase in R , revealed a more nuanced picture.

Speed, rather than body mass, was associated with a decrease in r at midstance (**Suppl. Fig. 3a,b,h**; **Suppl. Table 5**). A kangaroo travelling 1 m s^{-1} faster would decrease r by approximately 4.2%. The muscle moment arm reduced with speed for the full range of speeds we measured ($1.99 - 4.48 \text{ m s}^{-1}$), with the maximum r occurring when the ankle was at $114.4 \pm 0.8^\circ$

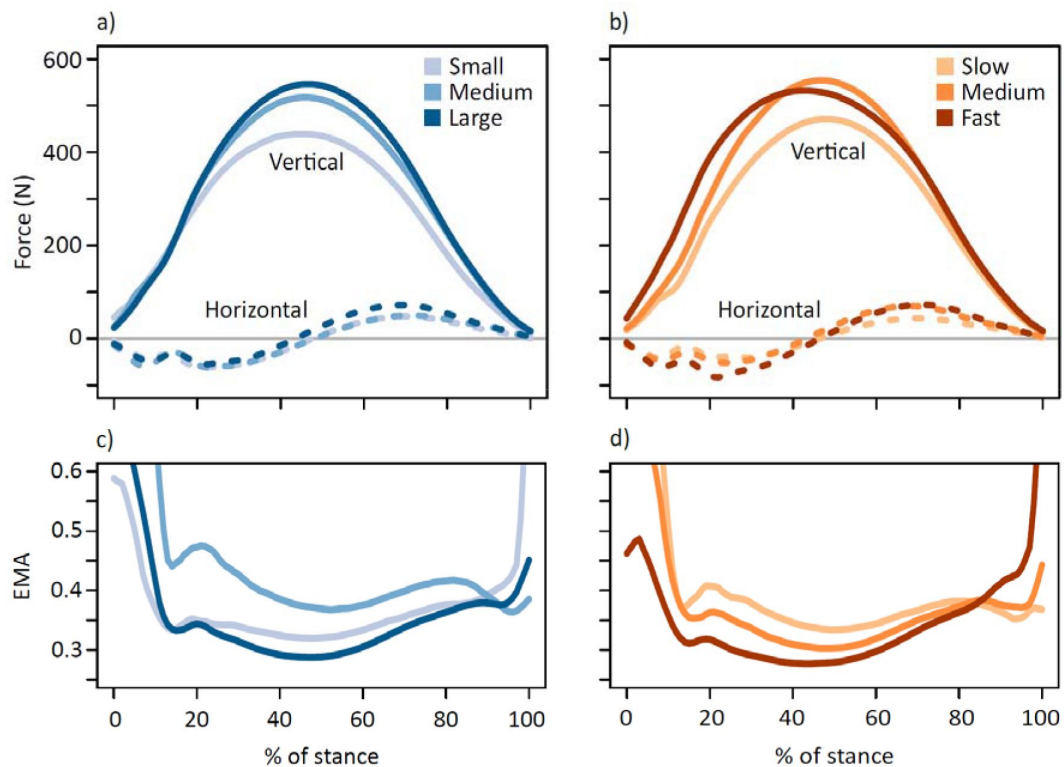


Figure 2

Horizontal fore-aft (dashed lines) and vertical (solid lines) components of the ground reaction force (GRF) (a) coloured by body mass subsets (small 17.6 ± 2.96 kg, medium 21.5 ± 0.74 kg, large 24.0 ± 1.46 kg) and (b) coloured by speed subsets (slow 2.52 ± 0.25 m s⁻¹, medium 3.11 ± 0.16 m s⁻¹, fast 3.79 ± 0.27 m s⁻¹). In (a) and (b) the medial-lateral component of the GRF is not shown as it remained close to zero, as expected for animals moving in a straight-line path. Lower panels show average time-varying EMA for the ankle joint subset by (c) body mass and (d) speed.

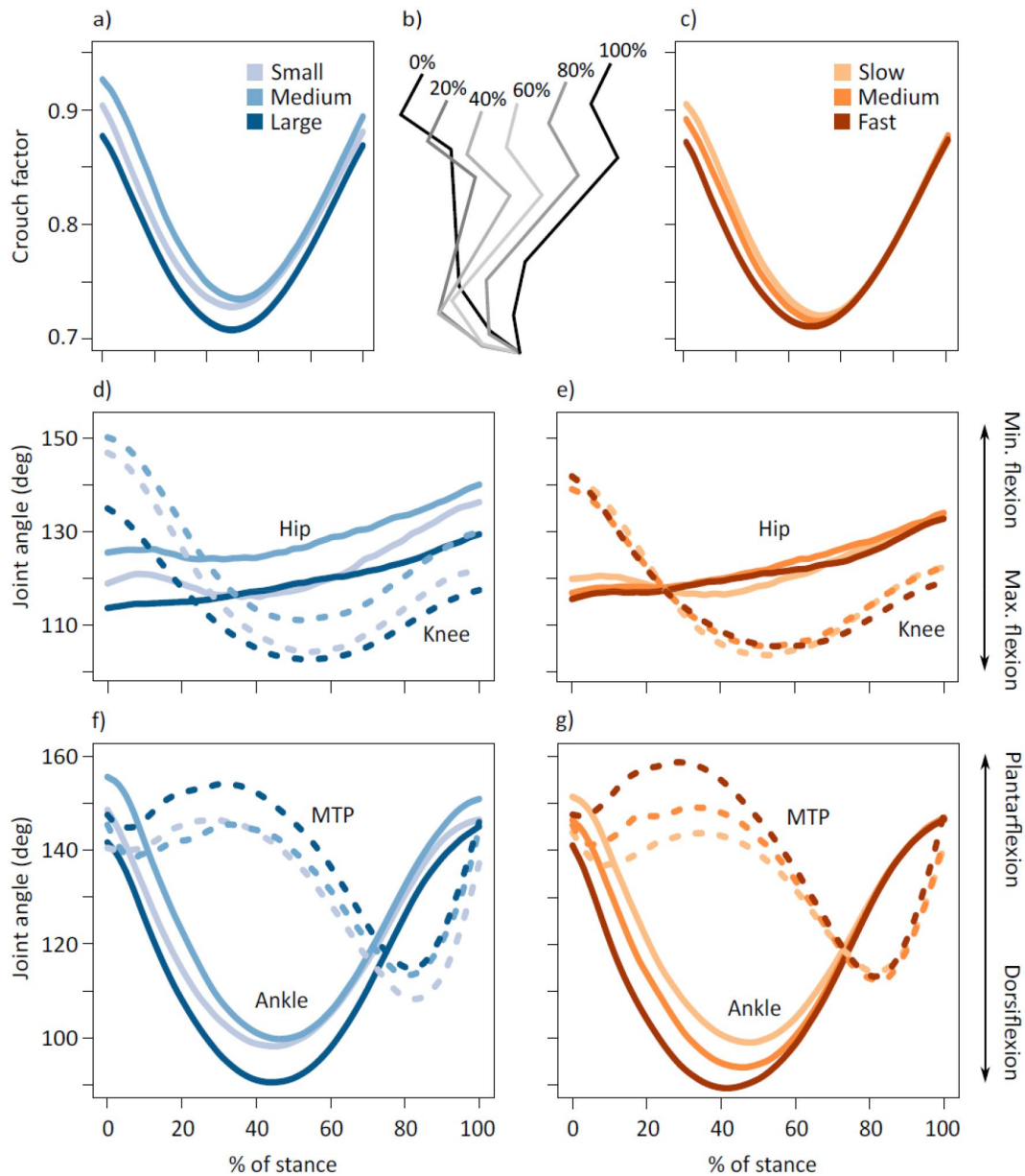


Figure 3

Average time-varying crouch factor (see [Fig. 1a](#)) of the kangaroo hindlimb grouped by (a) body mass and (c) speed. Position of the limb segments during % stance intervals (b). Average time-varying joint angles for the hip (solid lines) and knee (dashed lines) displayed for kangaroos grouped by (d) body mass and (e) speed. Average time-varying joint angles for the ankle (solid lines) and metatarsophalangeal (MTP) joints (dashed lines) displayed for kangaroos grouped by (f) body mass and (g) speed. For (f-g), increased plantarflexion represents a decrease in joint flexion, while increased dorsiflexion represents increased flexion of the joint. Body mass subsets: small 17.6 ± 2.96 kg, medium 21.5 ± 0.74 kg, large 24.0 ± 1.46 kg. Speed subsets: slow 2.52 ± 0.25 m s⁻¹, medium 3.11 ± 0.16 m s⁻¹, fast 3.79 ± 0.27 m s⁻¹.

(**Suppl. Fig. 3g** [↗](#)). Maximum ankle dorsiflexion ranged from 114.5° in the slowest trials, to 75.8° in faster trials (**Fig. 3g** [↗](#)). The timing of the local minimum r at midstance coincided with the timing of peak ankle dorsiflexion.

We observed an increase in R with body mass, but not speed, at midstance (**Suppl. Fig. 3c,d,h** [↗](#); **Suppl. Table 5** [↗](#)). We expected R to vary with both mass and speed because the increase in MTP plantarflexion prior to midstance reduced the distance between the ankle joint and the ground, which would increase R since the centre of pressure (CoP) at midstance remained in the same position (**Suppl. Table 2** [↗](#)). Increasing body mass by 4.6 kg or speed by 1 m s⁻¹ resulted in a ~20% reduction in ankle height (vertical distance from the ground) (**Suppl. Fig. 3e,f** [↗](#)). If we consider the nonsignificant relationship between R (and EMA) and speed to indicate that there is no change in R , then it conflicts with the ankle height and CoP result. Taking both into account, we think it is more likely that there is a small, but important, change in R , rather than no change in R with speed. It may be undetectable because we expect small effect sizes compared to the measurement range and measurement error (**Suppl. Fig. 3h** [↗](#)), or be obscured by a similar change in R with body mass. R is highly dependent on the length of the metatarsal segment, which is longer in larger kangaroos (1 kg BM corresponded to ~1% longer segment, $P < 0.001$, $R^2 = 0.449$). If R does indeed increase with speed, both R and r will tend to decrease EMA at faster speeds.

Joint moments

The change in the Achilles muscle moment arm, r , with speed has further implications for Achilles tendon stress, because force in the tendon is determined by the ratio of the ankle moment to r . As such, the decrease in r with speed would tend to increase tendon force (and thereby tendon stress), as would an increase in ankle moment. We found that speed, rather than mass, was associated with an increase in the maximum ankle moment (**Suppl. Fig. 4** [↗](#); **Suppl. Table 4** [↗](#), see supplementary material for details on other hindlimb joint moments). The range of body masses may not be sufficient to detect an effect of mass on ankle moment in addition to the effect of speed.

Tendon stress

Peak Achilles tendon stress increased as minimum EMA decreased (**Fig. 4a** [↗](#)). Subtle decreases in EMA which may have been undetected in previous studies correspond to discernible increases in tendon stress. For instance, reducing EMA from 0.242 (mean minimum EMA of the slow group) to 0.206 (mean minimum EMA of the fast group) was associated with an increase in tendon stress from ~50 MPa to ~60 MPa, decreasing safety factor from 2 to 1.67 (where 1 indicates failure), which is both measurable and physiologically significant. Tendon stress increased with body mass and speed (**Suppl. Table 5** [↗](#)). Maximum stress in the ankle extensor tendons occurred at 46.8±4.9% of stance, aligning with the timing of peak vertical GRF and maximum ankle moment. Body mass did not have a significant effect on the timing of peak stress, but peak stress occurred 3.2±0.8% earlier in stance with each 1 m s⁻¹ increase in speed ($P < 0.001$), reflecting the timing of the peak in ankle dorsiflexion.

The increase in tendon stress with speed, facilitated in part by the change in moment arms via the shift in posture, may explain changes in ankle work (c.f. Hypothesis (ii)).

Joint work and power

An analysis of joint-level energetics showed the majority of work and power per hop in the hindlimb was performed by the ankle joint (**Fig. 5** [↗](#); **Suppl. Table 8** [↗](#)). The ankle performed negative work in the braking period, as kinetic and gravitational potential energy was converted to elastic potential energy by loading the ankle extensor tendons. An increase in negative ankle

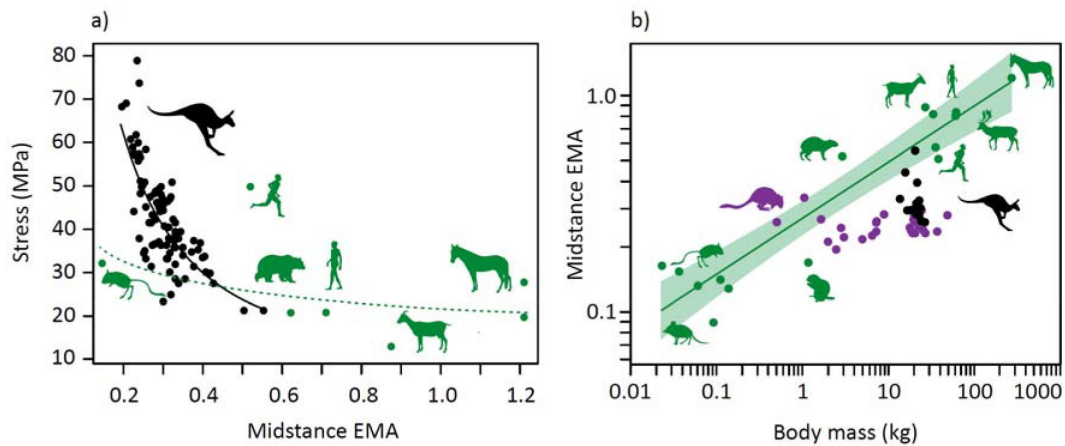


Figure 4

(a) Relationship between ankle effective mechanical advantage, EMA, at midstance and Achilles tendon stress (stress = $11.6 \text{ EMA}^{-1.04}$, $R^2 = 0.593$) (black), with other mammals (green). (b) Scaling of mean ankle EMA at midstance for each individual kangaroo against body mass (black), with data for a wider range of macropods (purple) (Bennett and Taylor 1995), and other mammals (green, $\text{EMA} = 0.269 \text{ M}^{0.259}$, shaded area 95% confidence interval) (Biewener 1990) shown.

work was associated with the increase in tendon stress in the ankle extensors ($B=0.750$, $SE=0.074$, $P<0.001$, $R^2=0.511$). Simple linear regression indicated that the increase in negative work was associated with speed but not body mass (**Fig. 6a**; **Suppl. Fig. 5e,f**; **Suppl. Table 6**).

Elastic potential energy was returned to do positive work in the propulsive period as the ankle started extending prior to midstance. Positive ankle work increased with both mass and speed (**Fig. 6a**; **Suppl. Fig. 5e,f**; **Suppl. Table 6**), but since it increased at the same rate as negative work, critically, there was no change in net ankle work with speed (**Fig. 6b**; **Suppl. Fig. 6e,f**; **Suppl. Table 6**). Thus in support of hypothesis (ii), the increase in positive and negative ankle work may be due to the increase in tendon stress rather than additional muscle work.

Discussion

How does posture contribute to kangaroo energetics?

The cost of generating force hypothesis (Taylor *et al.* 1980) implies that as animals increase locomotor speed and decrease ground contact time, metabolic rate should increase (Kram and Taylor 1990). Macropods defy this trend (Dawson and Taylor 1973, Baudinette *et al.* 1992, Kram and Dawson 1998). It is likely that the use of their ankle extensor tendons set them apart from other mammals, but the underlying mechanisms are unclear (Bennett and Taylor 1995, Bennett 2000, Thornton *et al.* 2022). We hypothesised, and found, that the hindlimb was more crouched at faster speeds, primarily due to the ankle and metatarsophalangeal joints. We propose that changes in moment arms (EMA) resulting from the change in posture would contribute to the increase in tendon stress with speed, and may thereby contribute to energetic savings by increasing the amount of positive and negative work done by the ankle, without requiring additional muscle work (**Fig. 7**).

Achilles tendon stress depends on extensor muscle force, which in turn depends on the muscle moment arm, ankle moment, external moment arm, and ground reaction force (GRF), assuming tendon material properties and cross-sectional area remain unchanged (**Fig. 7**). We detected an increase in peak ankle moment with speed independent of increases in body mass, despite the change in the external moment arm, R , with mass, likely due to a dominant effect of the GRF (ankle moment = $GRF \cdot R$). Peak GRF also naturally increased with speed together with shorter ground contact durations (**Fig. 2b**, **Suppl. Fig. 1b**). However, both the increase in ankle moment and decrease in the internal moment arm, r , with speed tend to increase tendon stress (stress = tendon force / cross-sectional area; tendon force = ankle moment / r). Although the increase in peak GRF with speed explains much of the increase in tendon stress (and strain) (**Suppl. Fig. 8**), the significant relationship between tendon stress and ankle EMA suggests that the increase in stress is not solely due to the greater GRFs, but that stress is also modulated by changes in posture throughout the stride (**Fig. 4a**). If peak GRF were constant but EMA changed from the average value of a slow hop to a fast hop, then stress would increase 18%, whereas if EMA remained constant and GRF varied by the same principles, then stress would only increase by 12%. Thus, changing posture and decreasing ground contact duration both appear to influence tendon stress for kangaroos, at least for the range of speeds we examined.

Previously, macropods were reported to have consistent ankle EMA due to the muscle moment arm, r , and the external moment arm, R , scaling similarly with body mass (**Fig. 4b**) (Bennett and Taylor 1995). Our results, however, suggest posture may not be as consistent as previously reported; we found that EMA and moment arms varied with both mass and speed. Larger and faster kangaroos were more crouched, leading to smaller ankle EMA. We were able to explore the individual contributions to ankle EMA using our musculoskeletal model, to demonstrate that changes in the ankle and metatarsophalangeal (MTP) range of motion resulted in changes in r with speed, and R with mass, respectively. R (and therefore also EMA) may also increase with

Figure 5

Average time-varying joint powers for the hip (a,b), knee (c,d), ankle (e,f), and MTP (g,h) displayed for kangaroos grouped by body mass (a, c, e, g) and speed (b, d, f, h). Power in all panels is set to the same scale. Body mass subsets: small 17.6 ± 2.96 kg, medium 21.5 ± 0.74 kg, large 24.0 ± 1.46 kg. Speed subsets: slow 2.52 ± 0.25 m s⁻¹, medium 3.11 ± 0.16 m s⁻¹, fast 3.79 ± 0.27 m s⁻¹.

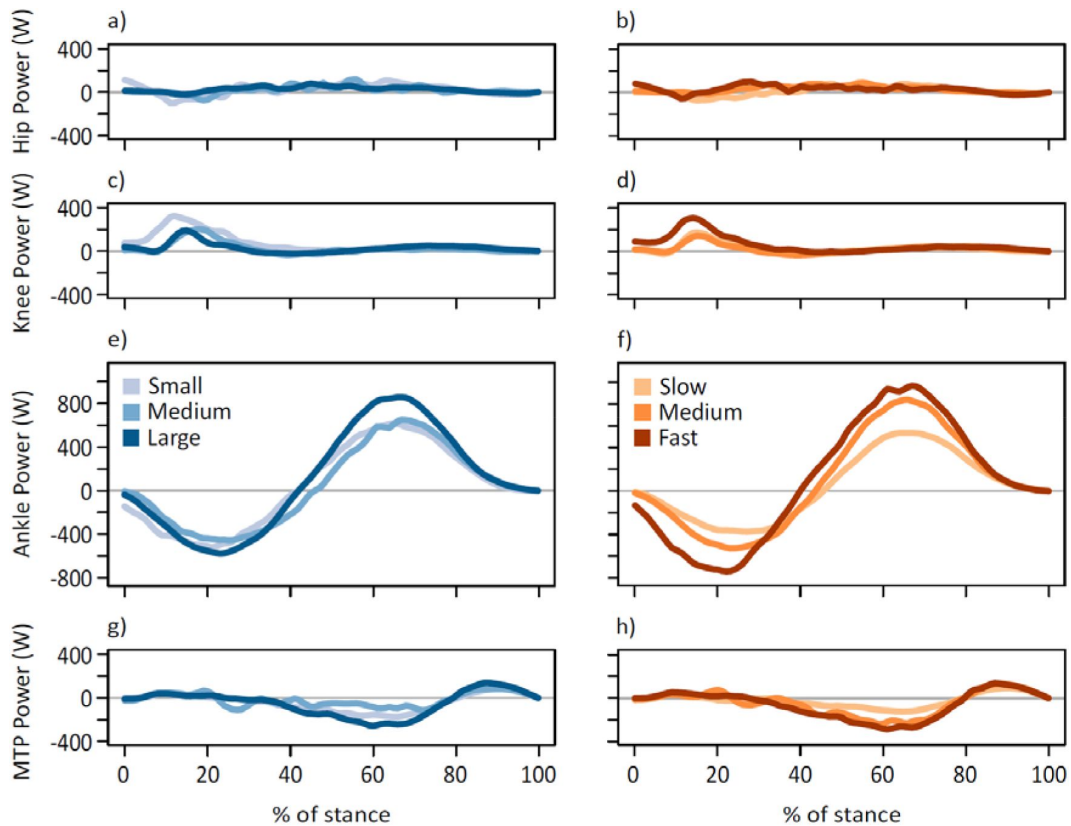
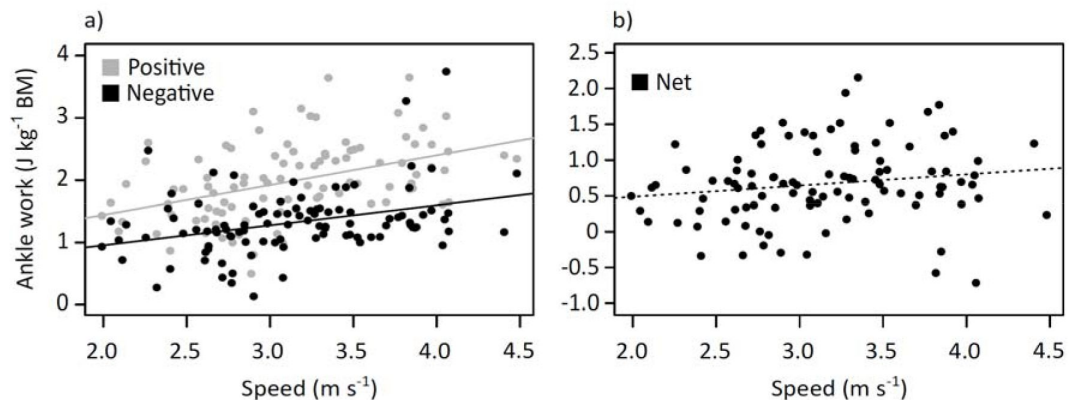


Figure 6

Variation with speed of (a) positive and negative ankle work, and (b) net ankle work per hop.



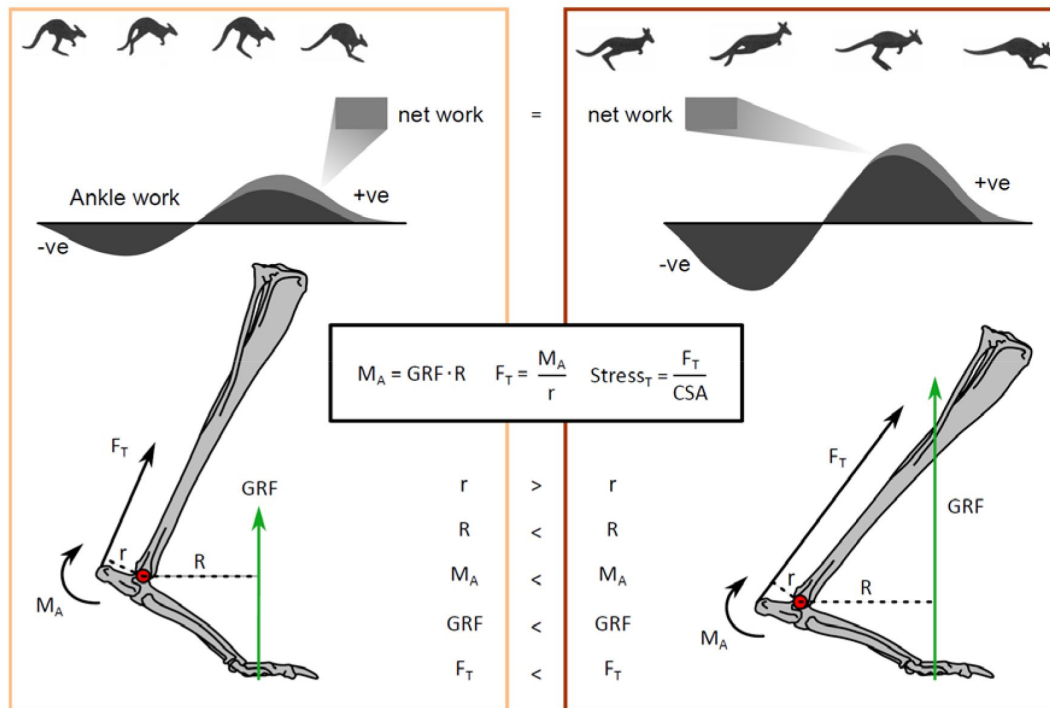


Figure 7

How the relationship between posture and speed is proposed to change tendon stress.

Forces are not to scale and joint angles are exaggerated for illustrative clarity. A slow hop (left panel) compared to a fast hop (right panel). The increase in ground reaction force (GRF) with speed, while a more crouched posture changes the muscle moment arm, r , and external moment arm, R , which allows the ankle to do more negative work (storing elastic potential energy in the tendons due to greater tendon stresses), without increasing net work, and thereby metabolic cost. Ankle moment was calculated by OpenSim and includes inertial terms.

speed, given that the ankle joint lowers substantially closer to the ground. These effects of the moment arms were additive; combined, there was a marked decrease in ankle EMA, particularly during the early stance when the ankle extensor tendons are loaded (**Fig. 2c,d**).

Given that tendon stress seems to be partially controlled by EMA, kangaroos could change their posture to absorb and return greater amounts of elastic strain energy, to meet the requirements of increasing positive ankle work at faster speeds. A comparison among all the hindlimb joints suggests the ankle was primarily responsible for both the storage and release of work during the stride (**Fig. 5**). Positive work increased with mass and speed, consistent with other studies in other species (Cavagna and Kaneko 1977). However, the amount of negative work absorbed during the stride also increased with speed, while, critically, net ankle work did not change, which indicates that the increase in negative work that was absorbed matched the increase in positive work that was required to move forward at more rapid speeds (**Fig. 6**). Indeed, we observed that all trials had a similar positive net ankle work (mean: $0.67 \pm 0.54 \text{ J kg}^{-1}$). The consistent net work observed among all speeds suggests that the ankle extensor muscle-tendon units were performing similar amounts of ankle work independent of speed, which would predominantly be done by the tendon. As such, the change in ankle EMA appears to be a mechanism that contributes to the increase in tendon stress and positive ankle work. This mechanism may help explain why hopping macropods do not follow the energetic trends observed in other species (Dawson and Taylor 1973, Baudinette *et al.* 1992, Kram and Dawson 1998), but it does not fully resolve the cost of generating force conundrum.

Decoupling force and energy during hopping

As Achilles tendon stress increases with speed in hopping kangaroos, so too must the muscle force in the gastrocnemius and plantaris, otherwise the tendon would lengthen the muscle rather than store elastic energy. These increases in force could be achieved via increases in muscle activation, but this would also increase muscle-level energy consumption. The question remains; whether there are physiologically plausible scenarios that could lead to an increase in muscle force without increasing energy use. Experimental data from hopping macropods confirm that tendon force and EMG can be decoupled, whereby greater EMG does not translate to larger forces or tendon stresses (Biewener *et al.* 2004b Table 2). Thus, the decoupling of energetics and speed might be explained by a muscle level change in mechanics, driven by either force-length or force-velocity relationships. Force-length effects might explain this result if muscles could operate at more favourable regions of their force-length relationship during tendon loading. The gastrocnemius and plantaris muscles of tammar wallabies do not appear to shorten considerably during hopping, but rather are suggested to act near-isometrically as a strut – however this near isometric behaviour only occurs in the second half of the stance phase (Biewener *et al.* 1998). This pattern in muscle fibre length does not vary with hopping speed, despite large increases in muscle tendon forces (Biewener *et al.* 1998).

Force-velocity effects may instead play an important role. During hopping, the gastrocnemius and plantaris activate prior to ground contact, and the majority of force development during the first half of stance is associated with muscle fibre stretch, such that the fibres undergo substantial active lengthening prior to their near-isometric behaviour (Biewener *et al.* 2004b Fig 3). Faster hopping speeds are also accompanied by decreased contact durations, suggesting that the rate of this muscle stretch likely increases with speed. This is supported with tendon force transducer (buckle) and kinematic data of kangaroos hopping at faster speeds (Griffiths 1989). Griffiths demonstrated that faster speeds lead to greater muscle stretch rates, the result of which enables rates of force to rise 10 times faster when compared to supramaximal isometric contractions in the same muscle. Experiments exploring single fibre energetics in frogs show that lengthening contractions are less costly compared to isometric (Linari *et al.* 2003), allowing the muscle to act more efficiently. Thus, the changes in posture associated with increases in speed may allow increases in active muscle fibre stretch rates during the first half of stance, supporting

increases in tendon stress, and thereby also energy storage, without increases in activation or metabolic cost. However, a detailed musculoskeletal model that incorporates individual muscle mechanical and energetic models is necessary to confirm this phenomenon.

Considerations and limitations

First, we believe it is more likely that the changes in moment arms and EMA can be attributed to speed rather than body mass, given the marked changes in joint angles and ankle height observed at faster hopping speeds. However, our sample included a relatively narrow range of body masses (13.7 to 26.6 kg) compared to the potential range (up to 80 kg), limiting our ability to entirely isolate the effects of speed from those of mass. Future work should examine a broader range of body sizes. Second, kangaroos studied here only hopped at relatively slow speeds, which bounds our estimates of EMA and tendon stress to a less critical region. As such, we were unable to assess tendon stress at fast speeds, where increased forces would reduce tendon safety factors closer to failure. A different experimental or modelling approach may be needed, as kangaroos in enclosures seem unwilling to hop faster over force plates. Finally, we did not determine whether the EMA of proximal hindlimb joints (which are more difficult to track via surface motion capture markers) remained constant with speed. Although the hip and knee contribute substantially less work than the ankle joint ([Fig. 5](#)), the majority of kangaroo skeletal muscle is located around these proximal joints. A change in EMA at the hip or knee could influence a larger muscle mass than at the ankle, potentially counteracting or enhancing energy savings in the ankle extensor muscle-tendon units. Further research is needed to understand how posture and muscles throughout the whole body contribute to kangaroo energetics.

Why are macropods unique?

No other mammals are known to achieve the same energetic feat as macropods, despite similar tendons or stride parameters ([Thornton *et al.* 2022](#)), but macropods are unique in other ways. Kangaroos operate at much lower ankle EMA values than other large mammalian species ([Fig. 4b](#)). Mammals >18 kg tend to operate with EMA values 0.5 to 1.2. The only mammals with comparable EMA values to kangaroos are rodents <1 kg ([Fig. 4b](#)) ([Biewener 1990](#)), such as kangaroo rats which have relatively thicker tendons that may be less suited for recovering elastic strain energy ([Biewener and Blickhan 1988](#)) (but see: [Christensen *et al.* \(2022\)](#)). [Fig. 4a](#) shows the non-linear relationship between tendon stress and EMA in kangaroos, quadrupeds and humans. The range of EMA estimates for other mammals suggest they operate in the region of this curve where large changes in EMA would only produce small changes in stress. This implies EMA modulation in other species is not as effective a mechanism to increase tendon stress with increased running speed as that observed in kangaroos.

Were macropods performance or size limited?

The morphology and hopping gait that make kangaroos supremely adapted for efficient locomotion likely also have several performance limitations. One possible consequence is a predicted reduction in manoeuvrability ([Biewener 2005](#)). The high compliance of the Achilles tendon would limit the ability to rapidly accelerate, owing to the lag between muscle force production and the transmission of this force to the environment. A second important limitation is the requirement for kangaroos to operate at large tendon stresses. Previous research has suggested that kangaroos locomote at dangerously low safety factors for tendon stress, predicted to be between 1-2 for large kangaroos ([Kram and Dawson 1998](#), [McGowan *et al.* 2008](#), [Snelling *et al.* 2017](#), [Thornton *et al.* 2022](#)). Our new insights into the mass and speed modulated changes in EMA suggest that these safety factors may be even lower, likely limiting the maximum body mass that hopping kangaroos can achieve. This suggests that previous projections of tendon stress may have overestimated the body mass at which tendons reach their safety limit ('safety factor' of 1). [Snelling *et al.* \(2017\)](#) and [McGowan *et al.* \(2008\)](#) estimated the maximum body mass to remain above this limit was approximately 150 kg, but even if we consider this is a

conservative prediction, it is far lower than the estimated mass of extinct macropodids (up to 240 kg (Helgen *et al.* 2006 [↗](#), Janis *et al.* 2023 [↗](#)). Thus we expect there must be a body mass where postural and gait changes shift from contributing to stress to mitigating it (Dick and Clemente 2017 [↗](#)).

This study highlights how EMA may be more dynamic than previously assumed, and how musculoskeletal modelling and simulation approaches can provide insights into direct links between form and function which are often challenging to determine from experiments alone.

Methods

Animals and data collection

Hopping data for red and eastern grey kangaroos was collected at Brisbane's Alma Park Zoo in Queensland, Australia, in accordance with approval by the Ethics and Welfare Committee of the Royal Veterinary College; approval number URN 2010 1051. The dataset includes 16 male and female kangaroos ranging in body mass from 13.7 to 26.6 kg (20.9 ± 3.4 kg). Two juvenile red (*Macropus rufus*) and 11 grey kangaroos (*Macropus giganteus*) were identified, while the remaining three could not be differentiated as either species. Body mass was determined in several ways, primarily by measurements of the kangaroos standing stationary on the force plate. If the individual did not stop on the force plate, and forward velocity was constant, then body mass was determined by dividing the total impulse across a constant velocity hop cycle (foot strike to foot strike) by the total hop cycle time and further dividing by gravitational acceleration. Finally, if neither of these approaches were sufficient, we interpolated from the relationship between leg marker distances (as proxy for segment lengths) and body mass. These methods produce estimates of body masses close to stationary measurements, where both were available.

Experimental protocol

Kangaroos hopped down a runway (~10×1.5 m) that was constructed in their enclosure using hessian cloths and stakes (Suppl. Video 1 [↗](#)). The kangaroos elected to hop between 1.99 and 4.48 m s⁻¹, with a range of speeds and number of trials for each individual (Suppl. Fig. 9 [↗](#)). The runway was open at both ends with two force plates (Kistler custom plate (60×60 cm) and AMTI Accugait plate (50×50 cm)) set sequentially in the centre and buried flush with the surface. The force plates recorded ground reaction forces (GRF) in the vertical, horizontal and lateral directions.

A 6-camera 3D motion capture system (Vicon T160 cameras), recorded by Nexus software (Vicon, Oxford, UK) at 200 Hz, was used to record kinematic data. Reflective markers were placed on the animals over the estimated hip, knee, ankle, metatarsophalangeal (MTP) joints; the distal end of phalanx IV; the anterior tip of the ilium; and the base of the tail (Fig. 1a [↗](#)). Force plate data was synchronously recorded at 1000 Hz via an analogue to digital board integrated with the Vicon system.

Data analysis

We calculated the stride length from the distance between the ankle or MTP marker coordinates at equivalent time points in the stride. The stance phase was defined as the period when the vertical GRF was greater than 2% of the peak GRF. We determined ground contact duration and total stride duration from the frame rate (200 Hz) and the number of frames from contact to take-off (contact duration) and contact to contact (stride duration), respectively, and calculated the stride frequency. Stride parameter results are detailed in the supplementary material.

In most trials, the stride before and after striking the force plate was visible, providing a total of 173 strides. If two strides were present in a trial, we took the average of the two strides for that trial.

Trials were excluded if only one foot landed on the force plate or if the feet did not land near-simultaneously, to give a total of 100 trials, with variety in the number of trials per kangaroo (**Suppl Fig. 9** [↗](#)). We did not include trials where the kangaroo started from or stopped on the force plate. We assumed GRF was equally shared between each leg and divided the vertical, horizontal and lateral forces in half, and calculated all results for one leg. We normalised GRF by body weight.

The fore-aft movement of the centre of pressure (CoP) was recorded by the force plate within the motion capture coordinate system (**Fig. 1b** [↗](#)). We assumed that the force was applied along phalanx IV and that there was no medial-lateral movement of the CoP. It was necessary to assume the CoP was fixed along the medial-lateral axis because when two feet landed on the force plate, the lateral forces on each foot were not recorded, and indeed should have cancelled if the forces were symmetrical (i.e. if the kangaroo was hopping in a straight path and one foot is not in front of the other). We only used symmetrical trials to ensure reliable measures of the anterior-posterior movement of the CoP.

We calculated the hopping velocity and acceleration of the kangaroo in each trial from the position of the pelvis marker, as this marker was close to the centre of mass and there should have been minimal skin movement artefact. Position data were smoothed using the ‘smooth.spline’ function in R (version 3.6.3). The average locomotor speed of the trial was taken as the mean horizontal component of the velocity during the aerial phase before and after the stance phase.

Building a kangaroo musculoskeletal model

We created a musculoskeletal model based on the morphology of a kangaroo for use in OpenSim (v3.3; [Seth et al. \(2018\)](#) [↗](#)) (**Fig. 1a** [↗](#)). The skeletal geometry was determined from a computed tomography (CT) scan of a mature western grey kangaroo (MorphoSource.org, Duke University, NC, USA). Western grey kangaroos are morphologically similar to eastern grey and red kangaroos ([Thornton et al. 2022](#) [↗](#)).

We extracted the skeletal components from the CT scan using Dragonfly (Version 2020.2, Object Research Systems (ORS) Inc., Montreal, Canada) and partitioned the hindlimb into five segments (pelvis, femur, tibia, metatarsals and calcaneus, and phalanges). The segments were imported into Blender (version 3.0.0, <https://www.blender.org/> [↗](#); Amsterdam, Netherlands) to clean and smooth the bones, align the vertebrae, and export the segments as meshes. We imported the meshes into Rhinoceros (version 6.0, Robert McNeel & Associates, Seattle, WA, USA) to construct the framework of the movement system.

All joints between segments were modelled as hinge joints which constrain motion to a single plane and one degree of freedom (DOF), except the hips which were modelled as ball-and-socket joints with three DOF. The joints were restricted to rotational (no translational) movement. The joints were marked with an origin (joint centre) and coordinate system determined by the movement of the segment (x is abduction/adduction, y is pronation/supination, z is extension/flexion). The limb bones and joints from one leg were mirrored about the sagittal plane to ensure bilateral symmetry.

The segment masses for a base model were determined from measurements of eastern grey and red kangaroos provided in [Hopwood \(1976\)](#) [↗](#). Cylinders set to the length and mass of each segment were used to approximate the segment centre of mass and moments of inertia.

We scaled the model to the size, shape and mass of each kangaroo using the OpenSim scale tool and kangaroo markers. A static posture was defined based on the 3D positions of the markers at midstance. Each segment was scaled separately, allowing for different scaling factors across segments but preserving mass distribution. The markers with less movement of the skin over the joint (e.g. the MTP marker) were more highly weighted than the markers with substantial skin movement (e.g. the hip and knee markers) in the scaling tool.

Joint kinematics and mechanics

We used inverse kinematics to determine time-varying joint angles during hopping. Inverse kinematics is an optimisation routine which simulates movement by aligning the model markers with the markers in the kinematic data for each time step ([Suppl. Video 2](#)). We adjusted the weighting on the markers based on the confidence and consistency in the marker position relative to the skeleton. The model movement from inverse kinematics was combined with GRFs in an inverse dynamics analysis in OpenSim to calculate net joint moments for the hip, knee, ankle and MTP joints throughout stance phase. Joint moments were normalised to body weight and leg length ([Fig. 1a](#)).

We calculated instantaneous joint powers (joint work) for each of the hindlimb joints over the stance phase of hopping as the product of joint moment (not normalised) and joint angular velocity. To determine joint work, we integrated joint powers with respect to time over discrete periods of positive and negative work, consistent with [Dick *et al.* \(2019\)](#). For the stance duration of each hop, all periods of positive work were summed and all periods of negative work were summed to determine the positive work, negative work, and net work done at each of the hindlimb joints for a hop cycle. Joint work was normalised to body mass.

Posture and EMA

We evaluated overall hindlimb posture to determine how crouched or upright the hindlimbs were during the stance phase of hopping. The total hindlimb length was determined as the sum of all segment lengths between the joint centres, from the toe to ilium ([Fig. 1a](#)). The crouch factor (CF) was calculated as the distance between the toe and the ilium marker divided by the total hindlimb length. Larger CF values indicated extended limbs whereas smaller CF values indicated more crouched postures.

We calculated the effective mechanical advantage (EMA) at the ankle as the muscle moment arm of the combined gastrocnemius and plantaris tendon, r , divided by the external moment arm, R (perpendicular distance between the GRF vector and the ankle joint) ([Fig. 1b](#)). We dissected (with approval from University of the Sunshine Coast Ethics Committee, ANE2284) a road-killed 27.6 kg male eastern grey kangaroo and used this, combined with published anatomy on the origin and insertion sites, to determine the ankle extensor muscle-tendon unit paths on the skeleton in OpenSim ([Bauschulte 1972](#), [Hopwood and Butterfield 1976](#), [Hopwood and Butterfield 1990](#)). The value r to the gastrocnemius and plantaris tendons for all possible ankle angles were determined from OpenSim and scaled to the size of each kangaroo, while R was calculated as the perpendicular distance between the ankle marker and the GRF vector at the CoP.

Tendon stress

Ankle extensor tendon forces were estimated as the time-varying ankle moment divided by the time varying Achilles tendon moment arm. The sum of the gastrocnemius and plantaris tendon cross-sectional areas were scaled to kangaroo body mass for each kangaroo by interpolating literature values ([Snelling *et al.* 2017](#)). Forces were divided by tendon cross-sectional area to calculate tendon stress. We excluded the third ankle extensor tendon (flexor digitorum longus),

which has a shorter muscle moment arm and which stores ~10% of strain energy of the ankle extensors in tammar wallabies as it is primarily involved in foot placement rather than energy storage (Biewener and Baudinette 1995 [↗](#)).

Statistics

We used multiple linear regression (lm function in R, v. 3.6.3, Vienna, Austria) to determine the effects of body mass and speed on the stride parameters, ground reaction forces, joint angles and CF, joint moments, joint work and power, and tendon stress. We considered the interaction of body mass and speed first, and removed the interaction term from the linear model if the interaction was not significant. Effects were considered significant at the $p < 0.05$ level. Each trial was treated individually, and species was not used as a factor in the analysis as there was no systematic difference in outcome measures between kangaroo species. The data was grouped into body mass (small 17.6 ± 2.96 kg, medium 21.5 ± 0.74 kg, large 24.0 ± 1.46 kg) and speed (slow 2.52 ± 0.25 m s⁻¹, medium 3.11 ± 0.16 m s⁻¹, fast 3.79 ± 0.27 m s⁻¹) subsets for display purposes only.

Supplementary Results

Stride parameters

Hopping speed ranged from 1.99 to 4.48 m s⁻¹. Larger kangaroos tended to hop at slightly faster speeds ($B = 0.048$, $SE = 0.018$, $P = 0.009$, $R^2 = 0.057$), and due to this weak relationship between body mass and speed, both variables were considered in multiple linear regression models to determine their relative effects on the outcome measures (see Table 1).

Faster speeds were associated with a greater magnitude of acceleration in the braking period of the stance phase, i.e. minimum horizontal acceleration and maximum vertical acceleration (**Suppl. Fig. 1a** [↗](#)). There was no significant relationship between body mass and acceleration; however, there was a significant interaction between body mass and speed on maximum vertical acceleration, whereby smaller kangaroos had a greater change in vertical acceleration between slower and faster hopping speeds than larger kangaroos (Table 1).

Body mass and speed had different effects on ground contact duration (Table 1). There was a slight increase in contact duration in larger kangaroos. A stronger, opposing relationship was found with speed, and as speed increased, contact duration decreased. The relatively tight correlation between contact duration and hopping speed ($R^2 = 0.73$) could prove useful for predicting speed when contact duration can be accurately measured. In **Suppl. Fig. 1b** [↗](#) we combined our data with red kangaroo data from Kram and Dawson (1998) [↗](#) to extend the predictive range of both studies.

Larger kangaroos hopped with longer strides and lower frequencies than smaller kangaroos (Table 1), and stride length also increased with speed (**Suppl. Fig. 1c** [↗](#)). We found a significant decrease in stride frequency with mass and an increase with speed, if we did not consider the interaction term (**Suppl. Fig. 1d** [↗](#)). However, a significant interaction between body mass and speed suggests that larger kangaroos relied more on increases in stride frequency to increase hopping speed compared to smaller kangaroos (Table 1).

Ground reaction forces

It is commonly assumed that the GRF is vertical and at a maximum at midstance or 50% of stance (Bennett and Taylor 1995 [↗](#), Kram and Dawson 1998 [↗](#), McGowan *et al.* 2008 [↗](#), Snelling *et al.* 2017 [↗](#)) but our results suggest it occurred earlier, at $42.9 \pm 26.9\%$ of stance, and the wide range suggests such assumptions should be used with caution.

Kinematics and posture

Kangaroos were maximally crouched at midstance, with crouch factor (CF) reaching a minimum at $50.1 \pm 4.2\%$ of stance. Crouch factor (CF) at initial ground contact decreased at faster speeds, although the limb was similarly flexed during midstance ($P=0.295$). Consequently, CF changed less at faster speeds than slower speeds.

Larger kangaroos had lower hip and knee ranges of motion (ROM) compared to smaller kangaroos (**Fig. 3d**; **Suppl. Table 3**). In the distal hindlimb, ankle ROM increased with body mass, largely owing to an increase in dorsiflexion at midstance (**Fig. 3f**; **Suppl. Table 3**). The ROM of the MTP joint did not change with body mass; however, there was both an increase in plantarflexion and a decrease in dorsiflexion, resulting in a shift to larger MTP angles with mass (**Fig. 3f**; **Suppl. Table 3**).

Joint moments

Despite the small changes in joint rotation in the proximal hindlimb, we detected a decrease in magnitude of the dimensionless hip extensor and knee flexor moments with mass (**Suppl. Fig. 4a**, **Suppl. Table 4**), and an increase in magnitude of both joint moments with speed during the braking period of stance (**Suppl. Fig. 4b**). Maximum hip extensor moment and knee flexor moment were significantly influenced by the interaction between body mass and speed, suggesting that larger kangaroos increased the magnitude of the moments at a faster rate with speed compared to smaller kangaroos.

Joint work and power

The MTP was the only joint which did predominantly negative work. The MTP did more negative work with speed, while net MTP work decreased with both mass and speed (**Suppl. Fig. 5g,h**; **Suppl. Fig. 6g,h**; **Suppl. Table 6**). The MTP transition from negative work to positive work at $\sim 80\%$ stance, as the back of the foot started to leave the ground. Conversely, the knee did almost no negative work, and the hip did very little (**Fig. 5**; **Suppl. Fig. 5**; **Suppl. Fig. 6**). Net work slightly increased with mass and speed in the hip, while at the knee, net work increased only with speed (**Suppl. Table 6**).

Supplementary Videos

Suppl Video 1

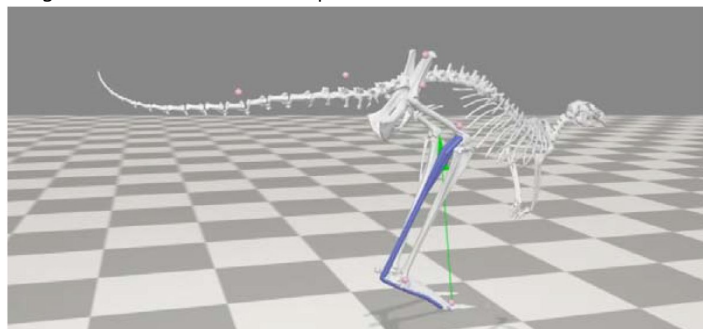
A red kangaroo hopping on the force plate during data collection.



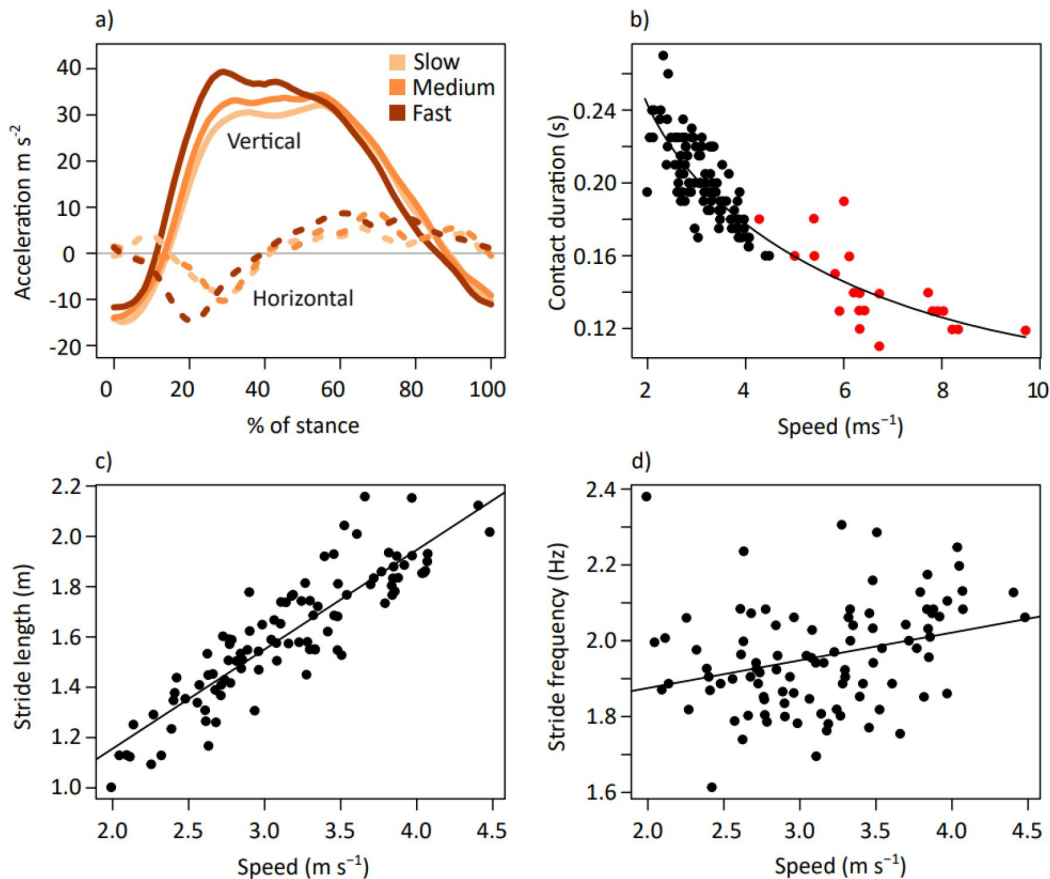
Suppl Video 2

Driving the musculoskeletal model with a recorded hopping trial.

The gastrocnemius and plantaris muscle-tendon unit (achilles) is shown in blue. The GRF appears in green. Markers associating the recorded kangaroos with the model are in pink.

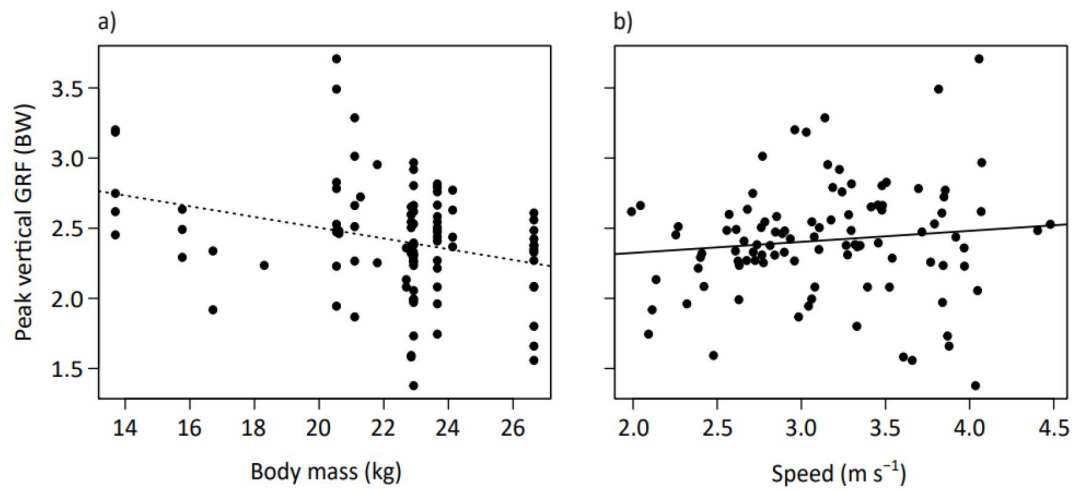


Supplementary Figures



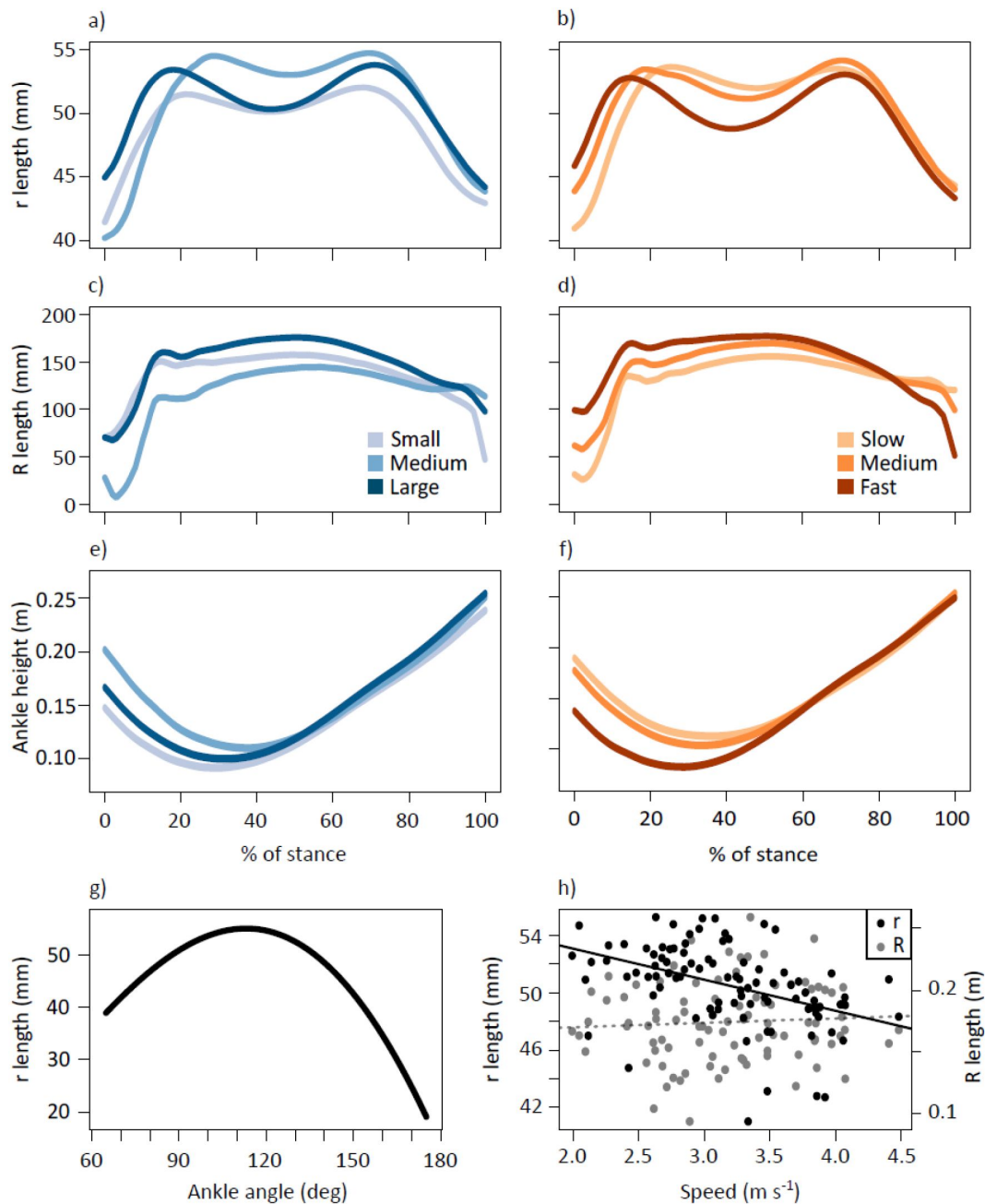
Suppl Figure 1

a) Mean vertical and horizontal components of whole body acceleration for kangaroos in the slow, medium and fast subsets (respectively: $2.52 \pm 0.25 \text{ m s}^{-1}$, $3.11 \pm 0.16 \text{ m s}^{-1}$, $3.79 \pm 0.27 \text{ m s}^{-1}$). (b) Ground contact duration across hopping speeds from current study (black circles) and for red kangaroos reported in Kram and Dawson (1998) (red circles). Regression equation: $t_c = 0.342 \text{ speed}^{-0.477}$ where t_c is contact duration and s is hopping speed. (c) Relationship between stride length and speed, and (d) stride frequency and speed.



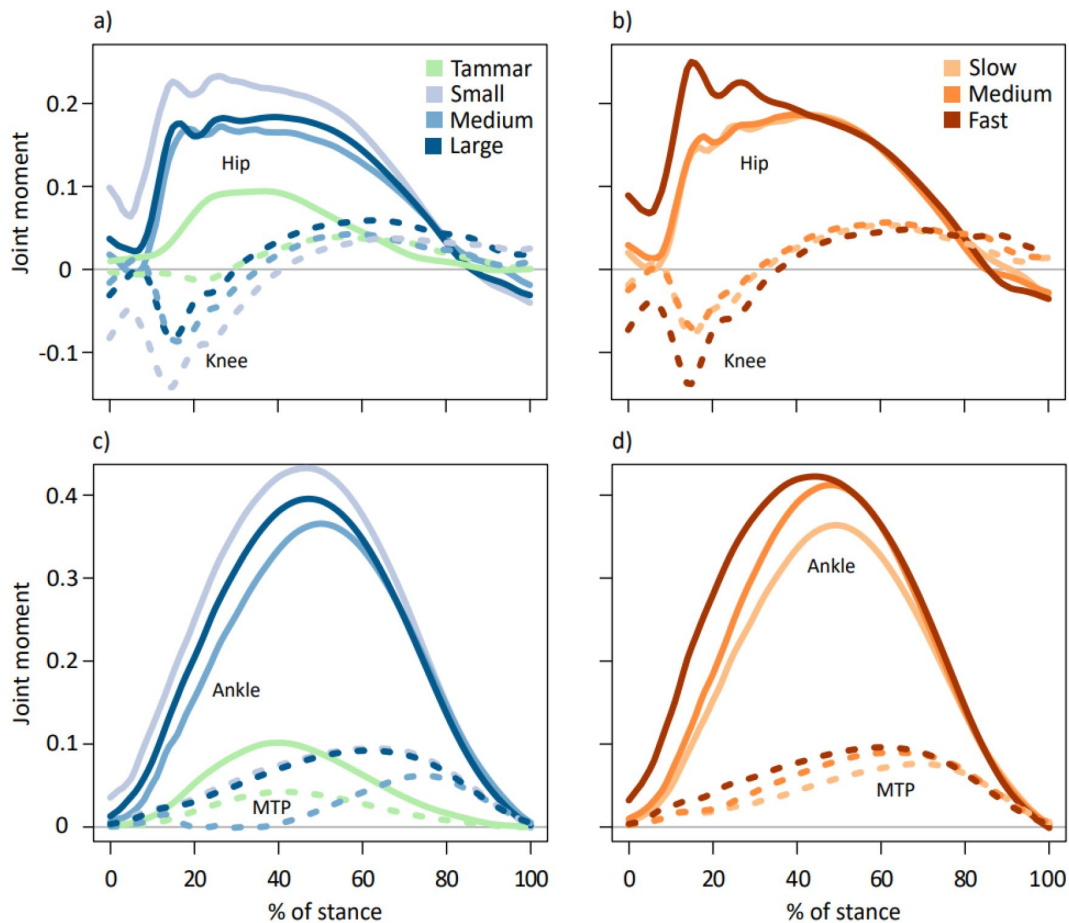
Suppl Figure 2

(a) Relationship between peak vertical GRF as a multiple of body weight (BW) with body mass and (b) with speed. Dotted line is insignificant and solid line is significant, see Table 2 for interaction.



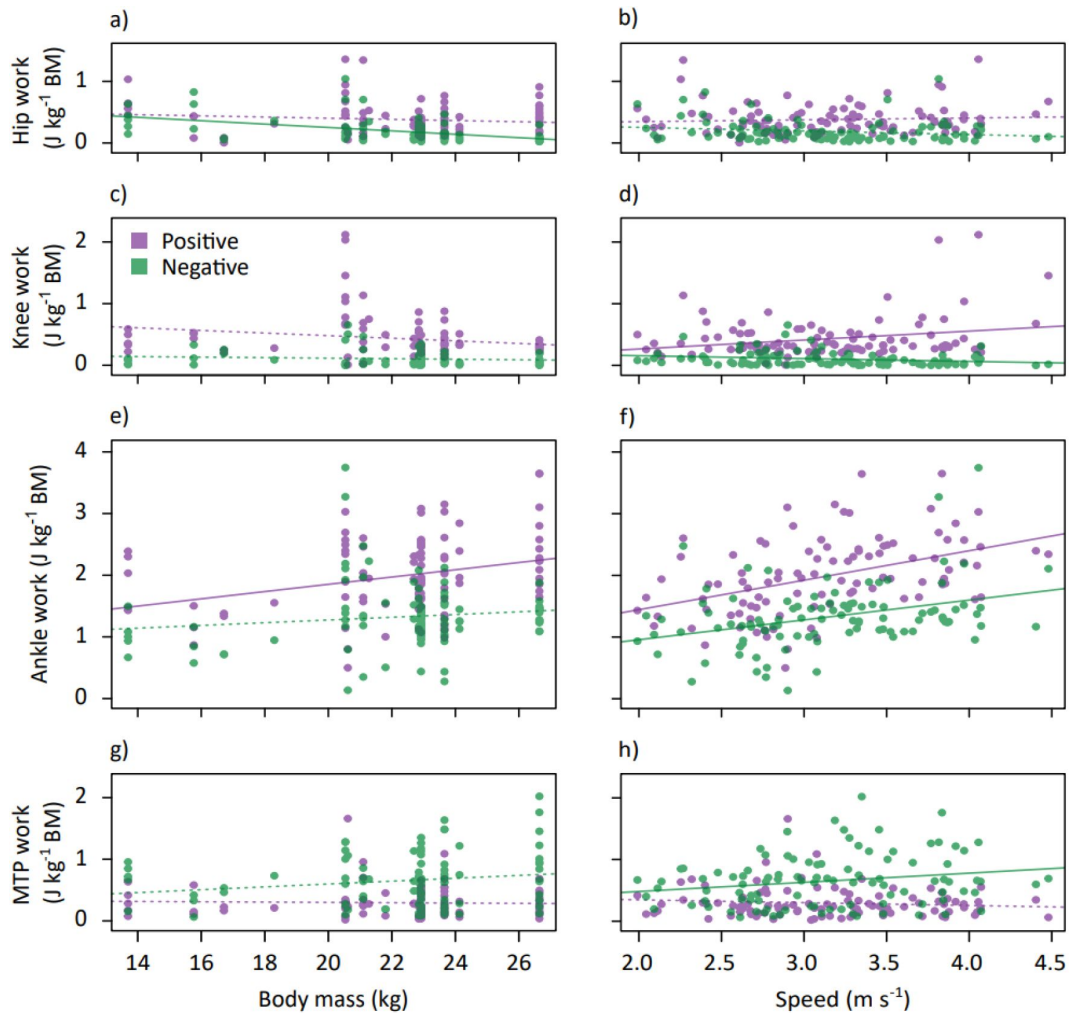
Suppl Figure 3

Average time-varying gastrocnemius and plantaris muscle moment arm, r , grouped by (a) body mass and (b) speed; external moment arm to the ankle, R , grouped by (c) body mass and (d) speed. Vertical displacement of the ankle marker from the ground throughout stance grouped by (e) body mass and (f) speed. Ankle angles and corresponding r arm length (g). Length of the r and R moment arms at midstance at different speeds (h). Body mass subsets: small 17.6 ± 2.96 kg, medium 21.5 ± 0.74 kg, large 24.0 ± 1.46 kg. Speed subsets: slow 2.52 ± 0.25 m s^{-1} , medium 3.11 ± 0.16 m s^{-1} , fast 3.79 ± 0.27 m s^{-1} .



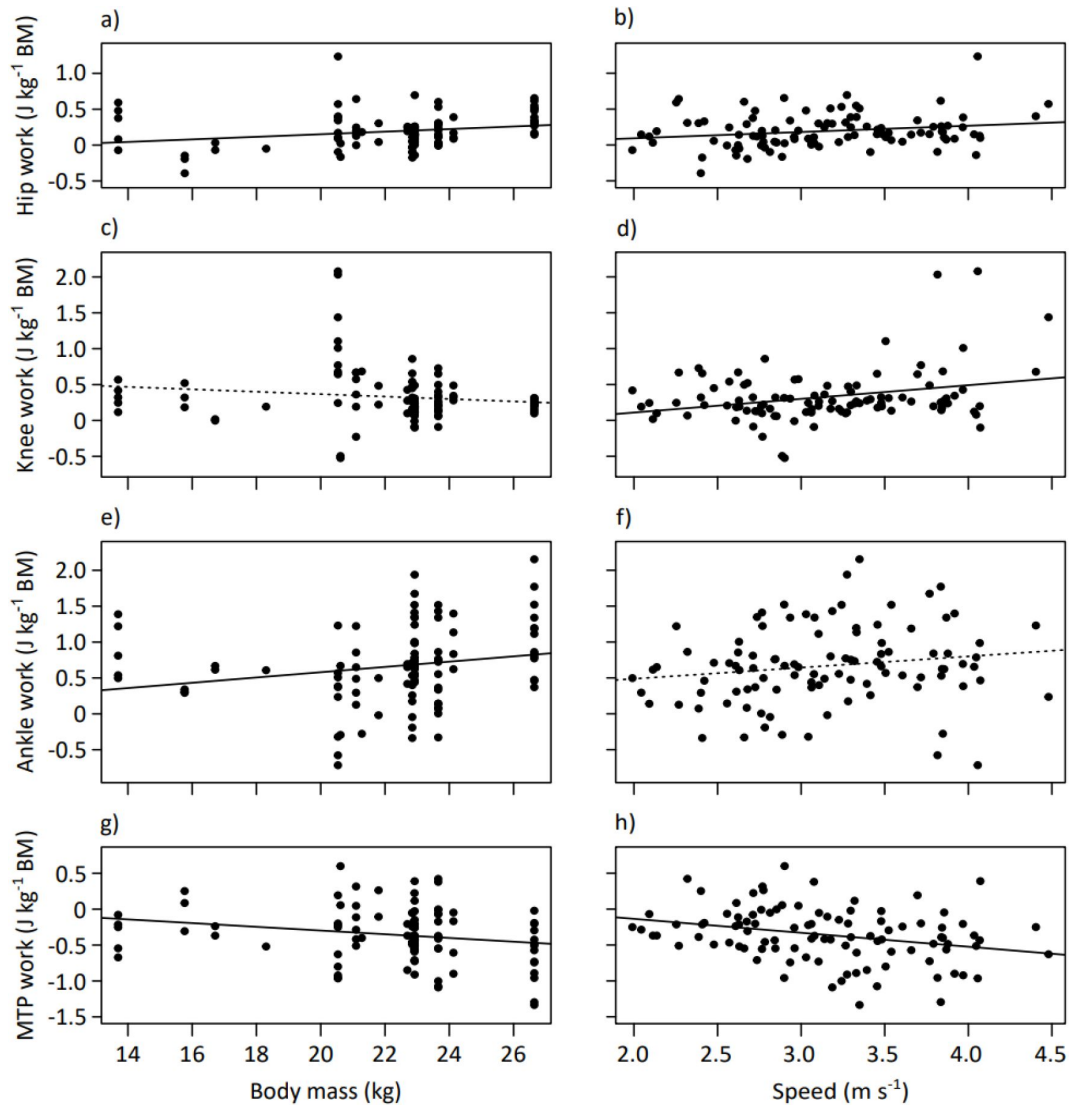
Suppl Figure 4

Average time-varying net joint moments (dimensionless, as moments were divided by body weight * leg length) for the hip (solid lines) and knee (dotted lines) displayed for kangaroos grouped by (a) body mass and (b) speed. Average time-varying net joint moments (dimensionless) for the ankle (solid lines) and metatarsophalangeal (MTP; dotted lines) joints displayed for kangaroos grouped by (c) body mass and (d) speed. Data for tammar wallabies was also included (McGowan *et al.* 2005 [\[link\]](#)) in green. Peak ankle moment occurred at 47.37 ± 4.91 % of the stance phase. Positive values represent extensor moments and negative values represent flexor moments. Body mass subsets: small 17.6 ± 2.96 kg, medium 21.5 ± 0.74 kg, large 24.0 ± 1.46 kg. Speed subsets: slow 2.52 ± 0.25 m s⁻¹, medium 3.11 ± 0.16 m s⁻¹, fast 3.79 ± 0.27 m s⁻¹.



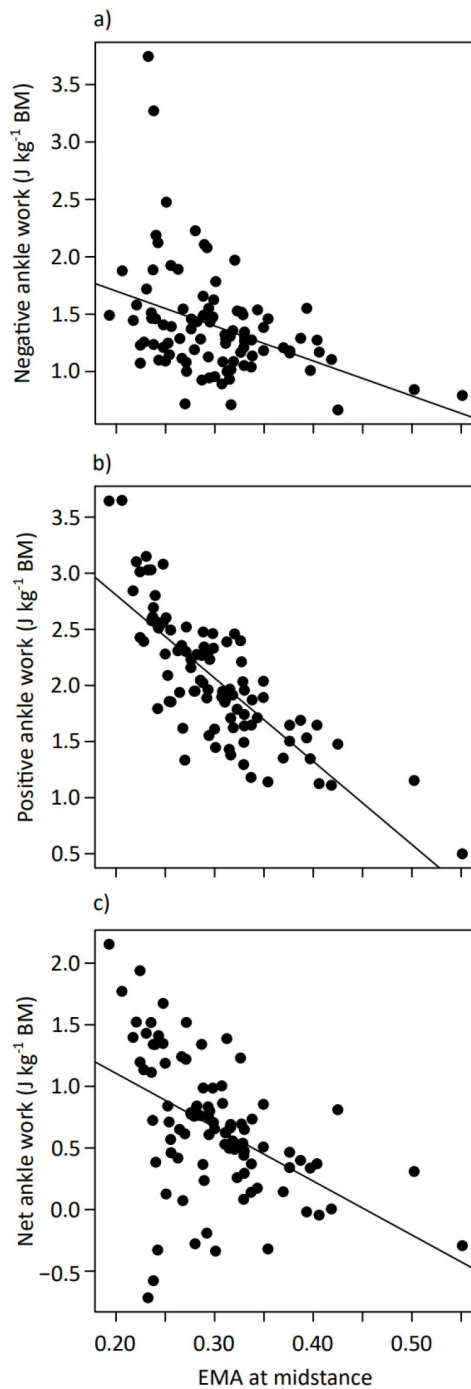
Suppl Figure 5

Positive (purple) and negative (green) joint work over stance for the hip, knee, ankle and MTP plotted against body mass (a,c,e,g) and speed (b,d,f,h). Solid lines represent significant trends, dotted lines are not significant (see [Suppl Table 6](#)).



Suppl Figure 6

Net joint work for the hip, knee, ankle and MTP joint over stance plotted against body mass (a,c,e,g) and speed (b,d,f,h). Solid lines represent significant trends, dotted lines are not significant (see [Suppl Table 6](#)).

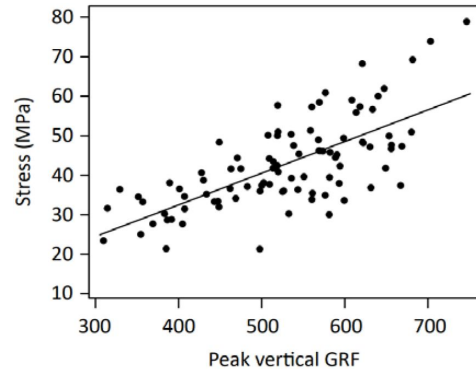


Suppl Figure 7

Negative (a) ($B=-3.04$, $SE=0.75$, $P<0.001$, $R^2=0.155$), positive (b) ($B=-7.42$, $SE=0.61$, $P<0.001$, $R^2=0.622$), and net ankle work (c) ($B=-4.37$, $SE=0.84$, $P<0.001$, $R^2=0.230$) plotted against EMA at 50% of stance.

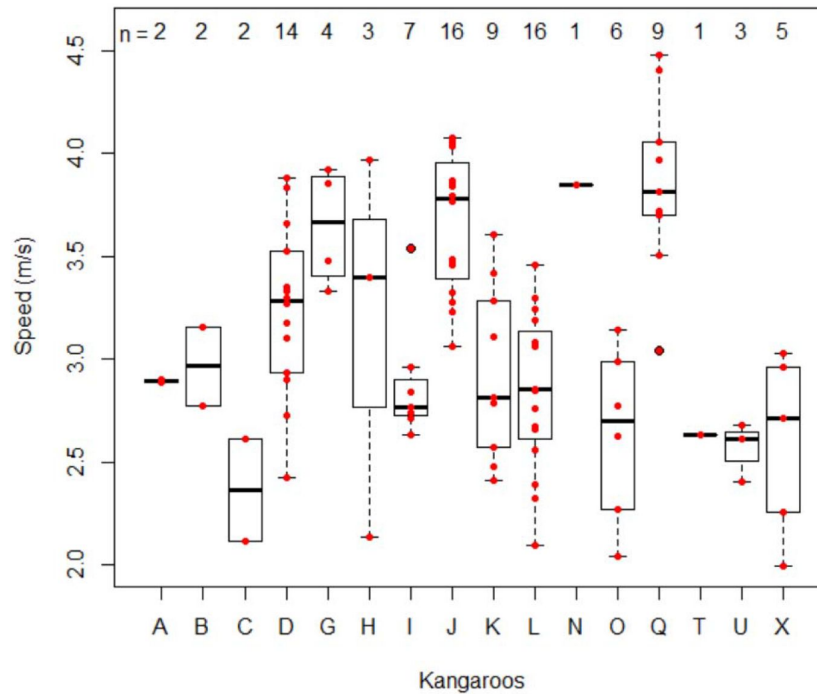
Suppl Figure 8

Peak vertical ground reaction force (GRF) plotted against tendon stress ($B=0.080$, $SE=0.009$, $P<0.001$, $R^2=0.486$).



Suppl Figure 9

Distribution of trial speeds and number of trials (n) per kangaroo (6.25 ± 5.02 trials per kangaroo).



Supplementary Tables

	Interaction			Body mass			Speed			R ²
	β	SE	P	β	SE	P	β	SE	P	
Maximum vertical acceleration (m s ⁻²)	-1.19	0.567	0.038	2.72	1.64	0.100	32.5	12.5	0.011	0.190
Maximum vertical acceleration (m s ⁻²)*				-0.673	0.294	0.024	6.32	1.60	0.000	0.152
Minimum vertical acceleration (m s ⁻²)				0.221	0.208	0.290	-0.268	1.13	0.814	0.012
Maximum horizontal acceleration (m s ⁻²)				-0.036	0.296	0.903	0.159	1.61	0.922	0.000
Minimum horizontal acceleration (m s ⁻²)				0.407	0.370	0.275	-6.05	2.02	0.003	0.087
Contact duration (ms)				2.87	0.393	0.000	-34.2	2.12	0.000	0.735
Stride length (m)				0.014	0.004	0.001	0.376	0.020	0.000	0.815
Stride frequency (Hz)	0.022	0.008	0.007	-0.082	0.023	0.001	-0.392	0.180	0.032	0.303
Stride frequency (Hz)*				-0.019	0.004	0.000	0.102	0.023	0.000	0.245

Suppl Table 1

Stride parameter multiple linear regression results as slopes, standard errors and P-values.

Models with a significant interaction are displayed in full, and as a simplified model without the interaction term included (marked *). The fit of the model is represented by R² and relationships are considered significant at P<0.05.

	Interaction			Body mass			Speed			R ²
	β	SE	P	β	SE	P	β	SE	P	
Normalised peak GRF (BW)	-0.055	0.024	0.026	0.111	0.070	0.117	1.34	0.535	0.014	0.172
Normalised peak GRF (BW)*				-0.045	0.013	0.001	0.142	0.068	0.040	0.128
Peak vertical GRF (N)	-11.5	5.31	0.032	47.9	15.4	0.002	281	117	0.019	0.332
Peak vertical GRF (N)*				15.0	2.76	0.000	28.4	14.9	0.060	0.299
Peak braking GRF (N)	4.87	2.04	0.019	-14.8	5.90	0.014	-130	45.0	0.005	0.218
Peak braking GRF (N)*				-0.917	1.07	0.392	-23.2	5.76	0.000	0.172
Peak propulsive GRF (N)				2.03	0.818	0.015	21.5	4.42	0.000	0.285
CoP location at midstance (mm)				-0.123	1.54	0.937	-15.2	8.34	0.071	0.036
CoP location at midstance corrected for phalanx size (mm)				-0.437	17.0	0.980	-62.7	91.9	0.497	0.005

represented by R² and relationships are considered significant at P<0.05.

Suppl Table 2

Ground reaction force and centre of pressure (CoP) multiple linear regression results as slopes, standard errors and P-values.

Models with a significant interaction are displayed in full, and as a simplified model without the interaction term included (marked *). The fit of the model is represented by R² and relationships are considered significant at P<0.05.

Suppl Table 3

Crouch factor (CF) and kinematics multiple linear regression results as slopes, standard errors and P-values.

Models with a significant interaction are displayed in full, and as a simplified model without the interaction term included (marked *). The fit of the model is represented by R^2 and relationships are considered significant at $P < 0.05$.

	Interaction			Body mass			Speed			R^2
	β	SE	P	β	SE	P	β	SE	P	
Maximum CF				-2.19	1.07	0.043	-15.8	5.74	0.007	0.122
Minimum CF				-2.45	0.885	0.007	-1.53	4.76	0.749	0.064
Change in CF				0.256	0.661	0.699	-14.2	3.55	0.000	0.129
Pelvis pitch ROM (deg)	0.935	0.298	0.002	-2.60	0.862	0.003	-19.9	6.58	0.003	0.100
Pelvis pitch ROM (deg)*				0.066	0.159	0.677	0.534	0.858	0.535	0.008
Hip ROM (deg)				-0.488	0.212	0.024	0.490	1.15	0.670	0.052
Minimum hip flexion (deg)				-0.655	0.315	0.040	0.703	1.70	0.681	0.043
Maximum hip flexion (deg)				-0.167	0.260	0.520	0.213	1.40	0.880	0.004
Knee ROM (deg)	-1.31	0.520	0.013	2.88	1.50	0.058	28.6	11.5	0.014	0.154
Knee ROM (deg)*				-0.850	0.272	0.002	-0.120	1.47	0.935	0.098
Minimum knee flexion (deg)	-1.72	0.649	0.010	3.76	1.88	0.048	39.5	14.3	0.007	0.162
Minimum knee flexion (deg)*				-1.13	0.34	0.001	1.95	1.84	0.294	0.101
Maximum knee flexion (deg)				-0.277	0.265	0.299	2.07	1.43	0.152	0.026
Ankle ROM (deg)				1.02	0.166	0.000	1.63	0.899	0.073	0.339
Peak ankle plantarflexion (deg)				-0.173	0.240	0.474	-3.87	1.30	0.004	0.104
Peak ankle dorsiflexion (deg)				-1.19	0.198	0.000	-5.49	1.07	0.000	0.466
MTP ROM (deg)				-0.098	0.259	0.707	7.16	1.40	0.000	0.219
Peak MTP plantarflexion (deg)				0.875	0.306	0.005	5.64	1.65	0.001	0.216
Peak MTP dorsiflexion (deg)				0.973	0.221	0.000	-1.52	1.20	0.205	0.166

Suppl Table 4

Multiple linear regression results of dimensionless peak joint moments as slopes, standard errors and P-values.

Models with a significant interaction are displayed in full, and as a simplified model without the interaction term included (marked *). The fit of the model is represented by R^2 and relationships are considered significant at $P < 0.05$.

	Interaction			Body mass			Speed			R^2
	β	SE	P	β	SE	P	β	SE	P	
Hip extensor moment	-0.020	0.006	0.001	0.049	0.018	0.007	0.525	0.136	0.000	0.268
Hip extensor moment*				-0.009	0.003	0.010	0.079	0.018	0.000	0.184
Knee extensor moment				0.000	0.001	0.766	0.003	0.008	0.721	0.003
Knee flexor moment	0.020	0.005	0.000	-0.048	0.015	0.002	-0.487	0.116	0.000	0.264
Knee flexor moment*				0.008	0.003	0.007	-0.060	0.015	0.000	0.158
Ankle extensor moment				-0.004	0.004	0.256	0.043	0.020	0.036	0.048
MTP extensor moment				-0.001	0.002	0.767	0.000	0.013	0.986	0.001

Suppl Table 5

Tendon stress and EMA multiple linear regression results as slopes, standard errors and P-values.

Models with a significant interaction are displayed in full, and as a simplified model without the interaction term included (marked *). The fit of the model is represented by R^2 and relationships are considered significant at $P < 0.05$.

	Interaction			Body mass			Speed			R^2
	β	SE	P	β	SE	P	β	SE	P	
<i>r</i> at midstance (mm)				-0.063	0.088	0.477	-1.88	0.474	0.000	0.173
<i>R</i> at midstance (mm)				3.49	1.03	0.001	-0.93	5.59	0.869	0.117
EMA at midstance				-6.64	1.98	0.001	-11.0	10.7	0.310	0.142
Peak tendon stress (MPa)				1.03	0.385	0.008	5.61	2.08	0.008	0.168
Normalised peak tendon stress				-0.048	0.018	0.008	0.258	0.096	0.008	0.107
Peak tendon stress timing %stance				-0.151	0.154	0.328	-3.13	0.831	0.000	0.159
Ankle height from ground (mm)				-0.298	0.706	0.674	-23.0	3.81	0.000	0.295

relationships are considered significant at $P < 0.05$.

Suppl Table 6

Joint net positive, negative and net work simple linear regression (lm(joint work ~ mass), lm(joint work ~ speed)) results as slopes, standard errors and P-values.

Work is normalised by body mass (BM). The fit of the model is represented by R^2 and relationships are considered

	Body mass				Speed			
	β	SE	P	R^2	β	SE	P	R^2
Hip pos	-0.010	0.008	0.238	0.014	0.030	0.044	0.497	0.005
Hip neg	-0.027	0.005	0.000	0.225	-0.058	0.031	0.066	0.034
Hip net	0.018	0.007	0.018	0.055	0.087	0.041	0.034	0.045
Knee pos	-0.021	0.011	0.052	0.038	0.145	0.057	0.013	0.061
Knee neg	-0.004	0.004	0.267	0.013	-0.046	0.020	0.026	0.050
Knee net	-0.017	0.012	0.169	0.019	0.191	0.064	0.003	0.084
Ankle pos	0.059	0.019	0.003	0.087	0.478	0.097	0.000	0.197
Ankle neg	0.022	0.017	0.200	0.017	0.321	0.087	0.000	0.122
Ankle net	0.037	0.017	0.037	0.044	0.156	0.095	0.102	0.027
MTP pos	-0.003	0.008	0.743	0.001	-0.046	0.043	0.286	0.012
MTP neg	0.023	0.014	0.089	0.029	0.149	0.073	0.044	0.041
MTP net	-0.026	0.012	0.035	0.045	-0.194	0.064	0.003	0.086

Suppl Table 7

Positive, negative and net joint work multiple linear regression results as slopes, standard errors and P-values.

Work is normalised by body mass (BM). Models with a significant interaction are displayed in full, and as a simplified model without the interaction term included (marked *). The fit of the model is represented by R^2 and relationships are considered significant at $P < 0.05$.

Joint work (J kg ⁻¹ BM)	Interaction			Body mass			Speed			R^2
	β	SE	P	β	SE	P	β	SE	P	
Hip positive				0.046	0.045	0.307	-0.012	0.008	0.161	0.025
Hip negative				-0.021	0.029	0.472	-0.026	0.005	0.000	0.229
Hip net				0.067	0.042	0.110	0.015	0.008	0.058	0.080
Knee positive	-0.072	0.020	0.000	1.766	0.431	0.000	0.175	0.057	0.003	0.241
Knee positive*				0.187	0.057	0.002	-0.030	0.011	0.006	0.133
Knee negative				-0.043	0.021	0.045	-0.002	0.004	0.570	0.053
Knee net	-0.073	0.022	0.002	1.818	0.491	0.000	0.179	0.064	0.007	0.219
Knee net*				0.230	0.064	0.001	-0.028	0.012	0.021	0.133
Ankle positive				0.424	0.099	0.000	0.039	0.018	0.038	0.232
Ankle negative	-0.080	0.032	0.015	2.052	0.707	0.005	0.234	0.093	0.013	0.176
Ankle negative*				0.311	0.091	0.001	0.007	0.017	0.671	0.123
Ankle net	0.122	0.033	0.000	-2.551	0.732	0.001	-0.315	0.096	0.001	0.173
Ankle net*				0.112	0.097	0.250	0.031	0.018	0.084	0.057
MTP positive				-0.045	0.044	0.312	0.000	0.008	0.956	0.012
MTP negative				0.125	0.075	0.101	0.017	0.014	0.217	0.056
MTP net				-0.170	0.066	0.011	-0.018	0.012	0.149	0.106

Suppl Table 8

The mean and standard deviation of joint work for all trials.

Positive, negative and net work is presented for each joint.

Joint work (J kg ⁻¹ BM)	Mean	SD
Hip positive	0.38	0.246
Hip negative	0.187	0.178
Hip net	0.193	0.235
Knee positive	0.432	0.334
Knee negative	0.104	0.117
Knee net	0.328	0.375
Ankle positive	1.99	0.613
Ankle negative	1.324	0.525
Ankle net	0.666	0.542
MTP positive	0.294	0.242
MTP negative	0.652	0.42
MTP net	-0.358	0.377

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Additional information

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Additional files

Suppl Figure 1 [↗](#)

Suppl Figure 2 [↗](#)

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Suppl Figure 5 [↗](#)

Suppl Figure 6 [↗](#)

Suppl Figure 7 [↗](#)

Suppl Figure 8 [↗](#)

Suppl Figure 9 [↗](#)

Suppl Video 1 [↗](#)

Suppl Video 2 [↗](#)

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Reviewer #1 (Public review):

Summary:

The study explored the biomechanics of kangaroo hopping across both speed and animal size to try and explain the unique and remarkable energetics of kangaroo locomotion.

Strengths:

Brings kangaroo locomotion biomechanics into the 21st century. Remarkably difficult project to accomplish. Excellent attention to detail. Clear writing and figures.

General Comments

This is a very impressive tour de force by an all-star collaborative team of researchers. The study represents a tremendous leap forward (pun intended) in terms of our understanding of kangaroo locomotion. Some might wonder why such an unusual species is of much interest. But, in my opinion, the classic study by Dawson and Taylor in 1973 of kangaroos launched the modern era of running biomechanics/energetics and applies to varying degrees to all animals that use bouncing gaits (running, trotting, galloping and of course hopping). The puzzling metabolic energetics findings of Dawson & Taylor (little if any increase in metabolic power despite increasing forward speed) remain a giant unsolved problem in comparative locomotor biomechanics and energetics. It is our "dark matter problem".

This study is certainly a hop towards solving the problem. The study clearly shows that the ankle and to a lesser extent the mtp joint are where the action is. They show in great detail by how much and by what means the ankle joint tendons experience increased stress at faster forward speeds. Since these were zoo animals, direct measures were not feasible, but the conclusion that the tendons are storing and returning more elastic energy per hop at faster speeds is solid. The conclusion that net muscle work per hop changes little from slow to fast forward speeds is also solid. Doing less muscle work can only be good if one is trying to minimize metabolic energy consumption. However, to achieve the greater tendon stresses, there must be greater muscle forces. Unless one is willing to reject the premise of the cost of generating force hypothesis, that is an important issue to confront. Further, the present data support the Kram & Dawson finding of decreased contact times at faster forward speeds. Kram & Taylor and subsequent applications of (and challenges to) their approach support the idea that shorter contact times (tc) require recruiting more expensive muscle fibers and hence greater metabolic costs. The present authors have clarified that this study has still not tied up the metabolic energetics across speed problem and they now point out how the group is now uniquely and enviably poised to explore the problem more using a dynamic SIMM model that incorporates muscle energetics.

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Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public Review):

Summary:

The study explored the biomechanics of kangaroo hopping across both speed and animal size to try and explain the unique and remarkable energetics of kangaroo locomotion.

Strengths:

The study brings kangaroo locomotion biomechanics into the 21st century. It is a remarkably difficult project to accomplish. There is excellent attention to detail, supported by clear writing and figures.

Weaknesses:

The authors oversell their findings, but the mystery still persists.

The manuscript lacks a big-picture summary with pointers to how one might resolve the big question.

General Comments

This is a very impressive tour de force by an all-star collaborative team of researchers. The study represents a tremendous leap forward (pun intended) in terms of our understanding of kangaroo locomotion. Some might wonder why such an unusual species is of much interest. But, in my opinion, the classic study by Dawson and Taylor in 1973 of kangaroos launched the modern era of running biomechanics/energetics and applies to varying degrees to all animals that use bouncing gaits (running, trotting, galloping and of course hopping). The puzzling metabolic energetics findings of Dawson & Taylor (little if any increase in metabolic power despite increasing forward speed)

remain a giant unsolved problem in comparative locomotor biomechanics and energetics. It is our "dark matter problem".

Thank you for the kind words.

This study is certainly a hop towards solving the problem. But, the title of the paper overpromises and the authors present little attempt to provide an overview of the remaining big issues.

We have modified the title to reflect this comment. "Postural adaptations may contribute to the unique locomotor energetics seen in hopping kangaroos"

The study clearly shows that the ankle and to a lesser extent the mtp joint are where the action is. They clearly show in great detail by how much and by what means the ankle joint tendons experience increased stress at faster forward speeds.

Since these were zoo animals, direct measures were not feasible, but the conclusion that the tendons are storing and returning more elastic energy per hop at faster speeds is solid. The conclusion that net muscle work per hop changes little from slow to fast forward speeds is also solid.

Doing less muscle work can only be good if one is trying to minimize metabolic energy consumption. However, to achieve greater tendon stresses, there must be greater muscle forces. Unless one is willing to reject the premise of the cost of generating force hypothesis, that is an important issue to confront. Further, the present data support the Kram & Dawson finding of decreased contact times at faster forward speeds. Kram & Taylor and subsequent applications of (and challenges to) their approach supports the idea that shorter contact times (tc) require recruiting more expensive muscle fibers and hence greater metabolic costs. Therefore, I think that it is incumbent on the present authors to clarify that this study has still not tied up the metabolic energetics across speed problems and placed a bow atop the package.

Fortunately, I am confident that the impressive collective brain power that comprises this author list can craft a paragraph or two that summarizes these ideas and points out how the group is now uniquely and enviably poised to explore the problem more using a dynamic SIMM model that incorporates muscle energetics (perhaps ala' Umberger et al.). Or perhaps they have other ideas about how they can really solve the problem.

You have raised important points, thank you for this feedback. We have added a limitations and considerations section to the discussion which highlights that there are still unanswered questions. Line 311-328

Considerations and limitations

"First, we believe it is more likely that the changes in moment arms and EMA can be attributed to speed rather than body mass, given the marked changes in joint angles and ankle height observed at faster hopping speeds. However, our sample included a relatively narrow range of body masses (13.7 to 26.6 kg) compared to the potential range (up to 80 kg), limiting our ability to entirely isolate the effects of speed from those of mass. Future work should examine a broader range of body sizes. Second, kangaroos studied here only hopped at relatively slow speeds, which bounds our estimates of EMA and tendon stress to a less critical region. As such, we were unable to assess tendon stress at fast speeds, where increased forces would reduce tendon safety factors closer to failure. A different experimental or modelling approach may be needed, as kangaroos in enclosures seem unwilling to hop faster over force plates. Finally, we did not determine whether the EMA of proximal hindlimb joints (which are more difficult to track via surface motion capture

markers) remained constant with speed. Although the hip and knee contribute substantially less work than the ankle joint (Fig. 4), the majority of kangaroo skeletal muscle is located around these proximal joints. A change in EMA at the hip or knee could influence a larger muscle mass than at the ankle, potentially counteracting or enhancing energy savings in the ankle extensor muscle-tendon units. Further research is needed to understand how posture and muscles throughout the whole body contribute to kangaroo energetics.”

Additionally, we added a line “Peak GRF also naturally increased with speed together with shorter ground contact durations (Fig. 2b, Suppl. Fig 1b)” (line 238) to highlight that we are not proposing that changes in EMA alone explain the full increase in tendon stress. Both GRF and EMA contribute substantially (almost equally) to stress, and we now give more equal discussion to both. For instance, we now also evaluate how much each contributes: “If peak GRF were constant but EMA changed from the average value of a slow hop to a fast hop, then stress would increase 18%, whereas if EMA remained constant and GRF varied by the same principles, then stress would only increase by 12%. Thus, changing posture and decreasing ground contact duration both appear to influence tendon stress for kangaroos, at least for the range of speeds we examined” (Line 245-249)

We have added a paragraph in the discussion acknowledging that the cost of generating force problem is not resolved by our work, concluding that “This mechanism may help explain why hopping macropods do not follow the energetic trends observed in other species (Dawson and Taylor 1973, Baudinette et al. 1992, Kram and Dawson 1998), but it does not fully resolve the cost of generating force conundrum” Line 274-276.

I have a few issues with the other half of this study (i.e. animal size effects). I would enjoy reading a new paragraph by these authors in the Discussion that considers the evolutionary origins and implications of such small safety factors. Surely, it would need to be speculative, but that's OK.

We appreciate this comment from the reviewer, however could not extend the study to discuss animal size effects because, as we now note in the results: “The range of body masses may not be sufficient to detect an effect of mass on ankle moment in addition to the effect of speed.” Line 193

Reviewer #2 (Public Review):

Summary

This is a fascinating topic that has intrigued scientists for decades. I applaud the authors for trying to tackle this enigma. In this manuscript, the authors primarily measured hopping biomechanics data from kangaroos and performed inverse dynamics.

While these biomechanical analyses were thorough and impressively incorporated collected anatomical data and an Opensim model, I'm afraid that they did not satisfactorily address how kangaroos can hop faster and not consume more metabolic energy, unique from other animals. Noticeably, the authors did not collect metabolic data nor did they model metabolic rates using their modelling framework. Instead, they performed a somewhat traditional inverse dynamics analysis from multiple animals hopping at a self-selected speed.

In the current study, we aimed to provide a joint-level explanation for the increases of tendon stress that are likely linked to metabolic energy consumption.

We have now included a limitations section in the manuscript (See response to Rev 1). We plan to expand upon muscle level energetics in the future with a more detailed musculoskeletal model.

Within these analyses, the authors largely focused on ankle EMA, discussing its potential importance (because it affects tendon stress, which affects tendon strain energy, which affects muscle mechanics) on the metabolic cost of hopping. However, EMA was roughly estimated (CoP was fixed to the foot, not measured) and did not detectably associate with hopping speed (see results). Yet, the authors interpret their EMA findings as though it systematically related with speed to explain their theory on how metabolic cost is unique in kangaroos vs. other animals

As noted in our methods, EMA was not calculated from a fixed centre of pressure (CoP). We did fix the medial-lateral position, owing to the fact that both feet contacted the force plate together, but the anteroposterior movement of the CoP was recorded by the force plate and thus allowed to move. We report the movement (or lack of movement) in our results. The anterior-posterior axis is the most relevant to lengthening or shortening the distance of the ‘out-lever’ R, and thereby EMA. It is necessary to assume fixed medial-lateral position because a single force trace and CoP is recorded when two feet land on the force plate. The mediolateral forces on each foot cancel out so there is no overall medial-lateral movement if the forces are symmetrical (e.g. if the kangaroo is hopping in a straight path and one foot is not in front of the other). We only used symmetrical trials so that the anterior-posterior movement of the CoP would be reliable. We have now added additional details into the text to clarify this

Indeed, the relationship between R and speed (and therefore EMA and speed) was not significant. However, the significant change in ankle height with speed, combined with no systematic change in COP at midstance, demonstrates that R would be greater at faster speeds. If we consider the nonsignificant relationship between R and speed to indicate that there is no change in R, then these two results conflict. We could not find a flaw in our methods, so instead concluded that the nonsignificant relationship between R and speed may be due to a small change in R being undetectable in our data. Taking both results into account, we believe it is more likely that there is a non-detectable change in R, rather than no change in R with speed, but we presented both results for transparency. We have added an additional section into the results to make this clearer (Line 177-185) “If we consider the nonsignificant relationship between R (and EMA) and speed to indicate that there is no change in R, then it conflicts with the ankle height and CoP result. Taking both into account, we think it is more likely that there is a small, but important, change in R, rather than no change in R with speed. It may be undetectable because we expect small effect sizes compared to the measurement range and measurement error (Suppl. Fig. 3h), or be obscured by a similar change in R with body mass. R is highly dependent on the length of the metatarsal segment, which is longer in larger kangaroos (1 kg BM corresponded to ~1% longer segment, $P < 0.001$, $R^2 = 0.449$). If R does indeed increase with speed, both R and r will tend to decrease EMA at faster speeds.”

These speed vs. biomechanics relationships were limited by comparisons across different animals hopping at different speeds and could have been strengthened using repeated measures design

There is significant variation in speed within individuals, not just between individuals. The preferred speed of kangaroos is 2-4.5 m/s, but most individuals showed a wide speed range within this. Eight of our 16 kangaroos had a maximum speed that was 1-2m/s faster than their slowest trial. Repeated measures of these eight individuals comprises 78 out of the 100 trials. It would be ideal to collect data across the full range of speeds for all individuals, but it is not feasible in this type of experimental setting. Interference with animals such as chasing is dangerous to kangaroos as they are prone to adverse reactions to stress. We have now added additional information about the chosen hopping speeds into the results and methods

sections to clarify this “The kangaroos elected to hop between 1.99 and 4.48 m s⁻¹, with a range of speeds and number of trials for each individual (Suppl. Fig. 9).” (Line 381-382)

There are also multiple inconsistencies between the authors' theory on how mechanics affect energetics and the cited literature, which leaves me somewhat confused and wanting more clarification and information on how mechanics and energetics relate

We thank the reviewer for this comment. Upon rereading we now understand the reviewers position, and have made substantial revisions to the introduction and discussion (See comments below)

My apologies for the less-than-favorable review, I think that this is a neat biomechanics study - but am unsure if it adds much to the literature on the topic of kangaroo hopping energetics in its current form.

Again we thank the reviewer for their time and appreciate their efforts to strengthen our manuscript.

Reviewer #3 (Public Review):

Summary:

The goal of this study is to understand how, unlike other mammals, kangaroos are able to increase hopping speed without a concomitant increase in metabolic cost. They use a biomechanical analysis of kangaroo hopping data across a range of speeds to investigate how posture, effective mechanical advantage, and tendon stress vary with speed and mass. The main finding is that a change in posture leads to increasing effective mechanical advantage with speed, which ultimately increases tendon elastic energy storage and returns via greater tendon strain. Thus kangaroos may be able to conserve energy with increasing speed by flexing more, which increases tendon strain.

Strengths:

The approach and effort invested into collecting this valuable dataset of kangaroo locomotion is impressive. The dataset alone is a valuable contribution.

Thank you!

Weaknesses:

Despite these strengths, I have concerns regarding the strength of the results and the overall clarity of the paper and methods used (which likely influences how convincingly the main results come across).

(1) The paper seems to hinge on the finding that EMA decreases with increasing speed and that this contributes significantly to greater tendon strain estimated with increasing speed. It is very difficult to be convinced by this result for a number of reasons:

It appears that kangaroos hopped at their preferred speed. Thus the variability observed is across individuals not within. Is this large enough of a range (either within or across subjects) to make conclusions about the effect of speed, without results being susceptible to differences between subjects?

Apologies, this was not clear in the manuscript. Kangaroos hopping at their preferred speed means we did not chase or startle them into high speeds to comply with ethics and enclosure limitations. Thus we did not record a wide range of speeds within the bounds of what

kangaroos are capable of in the wild (up to 12 m/s), but for the range we did measure (~2-4.5 m/s), there is a large amount of variation in hopping speed within each individual kangaroo. Out of 16 individuals, eight individuals had a difference of 1-2m/s between their slowest and fastest trials, and these kangaroos accounted for 78 out of 100 trials. Of the remainder, six individuals had three or fewer trials each, and two individuals had highly repeatable speeds (3 out of 4, and 6 out of 7 trials were within 0.5 m/s). We have now removed the terminology “preferred speed” e.g line 115. We have added additional information about the chosen hopping speeds into the results and methods, including an appendix figure “The kangaroos elected to hop between 1.99 and 4.48 m s⁻¹, with a range of speeds and number of trials for each individual (Suppl. Fig. 9).” (Line 381-382)

In the literature cited, what was the range of speeds measured, and was it within or between subjects?

For other literature, to our knowledge the highest speed measured is ~9.5m/s (see supplementary Fig1b) and there were multiple measures for several individuals (see methods Kram & Dawson 1998).

Assuming that there is a compelling relationship between EMA and velocity, how reasonable is it to extrapolate to the conclusion that this increases tendon strain and ultimately saves metabolic cost? They correlate EMA with tendon strain, but this would still not suggest a causal relationship (incidentally the p-value for the correlation is not reported).

The functions that underpin these results (e.g. moment = GRF*R) come from physical mechanics and geometry, rather than statistical correlations. Additionally, a p-value is not appropriate in the relationship between EMA and stress (rather than strain) because the relationship does not appear to be linear. We have made it clearer in the discussion that we are not proposing that entire change in stress is caused by changes in EMA, but that the increase in GRF that naturally occurs with speed will also explain some of the increase in stress, along with other potential mechanisms. The discussion has been extensively revised to reflect this.

Tendon strain could be increasing with ground reaction force, independent of EMA. Even if there is a correlation between strain and EMA, is it not a mathematical necessity in their model that all else being equal, tendon stress will increase as ema decreases? I may be missing something, but nonetheless, it would be helpful for the authors to clarify the strength of the evidence supporting their conclusions.

Yes, GRF also contributes to the increase in tendon stress in the mechanism we propose (Suppl. Fig. 8), see the formulas in Fig 6, and we have made this clearer in the revised discussion (see above comment). You are correct that mathematically stress is inversely proportional to EMA, which can be observed in Fig. 7a, and we did find that EMA decreases.

The statistical approach is not well-described. It is not clear what the form of the statistical model used was and whether the analysis treated each trial individually or grouped trials by the kangaroo. There is also no mention of how many trials per kangaroo, or the range of speeds (or masses) tested.

The methods include the statistical model with the variables that we used, as well as the kangaroo masses (13.7 to 26.6 kg, mean: 20.9 ± 3.4 kg). We did not have sufficient within individual sample size to use a linear mixed effect model including subject as a random factor, thus all trials were treated individually. We have included this information in the results section.

We have now moved the range of speeds from the supplementary material to the results and figure captions. We have added information on the number of trials per kangaroo to the methods, and added Suppl. Fig. 9 showing the distribution of speeds per kangaroo.

We did not group the data e.g. by using an average speed per individual for all their trials, or by comparing fast to slow groups for statistical analysis (the latter was only for display purposes in our figures, which we have now made clearer in the methods statistics section).

Related to this, there is no mention of how different speeds were obtained. It seems that kangaroos hopped at a self-selected pace, thus it appears that not much variation was observed. I appreciate the difficulty of conducting these experiments in a controlled manner, but this doesn't exempt the authors from providing the details of their approach.

Apologies, this was not clear in the manuscript. Kangaroos hopping at their preferred speed means we did not chase or startle them into high speeds to comply with ethics and enclosure limitations. Thus we did not record a wide range of speeds within the bounds of what kangaroos are capable of in the wild (up to 12 m/s). We have now removed the terminology “preferred speed” e.g. line 115. We have added additional information about the chosen hopping speeds into the results and methods, including an appendix figure (see above comment). (Line 381-382)

Some figures (Figure 2 for example) present means for one of three speeds, yet the speeds are not reported (except in the legend) nor how these bins were determined, nor how many trials or kangaroos fit in each bin. A similar comment applies to the mass categories. It would be more convincing if the authors plotted the main metrics vs. speed to illustrate the significant trends they are reporting.

Thank you for this comment. The bins are used only for display purposes and not within the statistical analysis. We have clarified this in the revised manuscript: “The data was grouped into body mass (small 17.6 ± 2.96 kg, medium 21.5 ± 0.74 kg, large 24.0 ± 1.46 kg) and speed (slow 2.52 ± 0.25 m s⁻¹, medium 3.11 ± 0.16 m s⁻¹, fast 3.79 ± 0.27 m s⁻¹) subsets for display purposes only”. (Line 495-497)

(2) The significance of the effects of mass is not clear. The introduction and abstract suggest that the paper is focused on the effect of speed, yet the effects of mass are reported throughout as well, without a clear understanding of the significance. This weakness is further exaggerated by the fact that the details of the subject masses are not reported.

Indeed, the primary aim of our study was to explore the influence of speed, given the uncoupling of energy from hopping speed in kangaroos. We included mass to ensure that the effects of speed were not driven by body mass (i.e.: that larger kangaroos hopped faster). Subject masses were reported in the first paragraph of the methods, albeit some were estimated as outlined in the same paragraph.

(3) The paper needs to be significantly re-written to better incorporate the methods into the results section. Since the results come before the methods, some of the methods must necessarily be described such that the study can be understood at some level without turning to the dedicated methods section. As written, it is very difficult to understand the basis of the approach, analysis, and metrics without turning to the methods.

The methods after the discussion is a requirement of the journal. We have incorporated some methods in the results where necessary but not too repetitive or disruptive, e.g. Fig. 1 caption,

and specifying we are only analysing EMA for the ankle joint

Reviewing Editor (Recommendations For The Authors):

Below is a list of specific recommendations that the authors could address to improve the eLife assessment:

(1) Based on the data presented and the fact that metabolic energy was not measured, the authors should temper their conclusions and statements throughout the manuscript regarding the link between speed and metabolic energy savings. We recommend adding text to the discussion summarizing the strengths and limitations of the evidence provided and suggesting future steps to more conclusively answer this mystery.

There is a significant body of work linking metabolic energy savings to measured increases in tendon stress in macropods. However, the purpose of this paper was to address the unanswered questions about why tendon stress increases. We found that stress did not only increase due to GRF increasing with speed as expected, but also due to novel postural changes which decreased EMA. In the revised manuscript, we have tempered our conclusions to make it clearer that it is not just EMA affecting stress, and added limitations throughout the manuscript (see response to Rev 1).

(2) To provide stronger evidence of a link between speed, mechanics, and metabolic savings the authors can consider estimating metabolic energy expenditure from their OpenSIM model. This is one suggestion, but the authors likely have other, possibly better ideas. Such a model should also be able to explain why the metabolic rate increases with speed during uphill hopping.

Extending the model to provide direct metabolic cost estimates will be the goal of a future paper, however the models does not have detailed muscle characteristics to do this in the formulation presented here. It would be a very large undertaking which is beyond the scope of the current manuscript. As per the comment above, the results of this paper are not reliant on metabolic performance.

(3) The authors attempt to relate the newly quantified hopping biomechanics to previously published metabolic data. However, all reviewers agree that the logic in many instances is not clear or contradictory. Could one potential explanation be that at slow speeds, forces and tendon strain are small, and thus muscle fascicle work is high? Then, with faster speeds, even though the cost of generating isometric force increases, this is offset by the reduction in the metabolic cost of muscular work. The paper could provide stronger support for their hypotheses with a much clearer explanation of how the kinematics relate to the mechanics and ultimately energy savings.

In response to the reviewers comments, we have substantially modified the discussion to provide clearer rationale.

(4) The methods and the effort expended to collect these data are impressive, but there are a number of underlying assumptions made that undermine the conclusions. This is due partly to the methods used, but also the paper's incomplete description of their methods. We provide a few examples below:

It would be helpful if the authors could speak to the effect of the limited speeds tested and between-animal comparisons on the ability to draw strong conclusions from the present dataset. ·

Throughout the discussion, the authors highlight the relationship between EMA and speed. However, this is misleading since there was no significant effect of speed on EMA. Speed only affected the muscle moment arm, r . At minimum, this should be clarified and the effect on EMA not be overstated. Additionally, the resulting implications on their ability to confidently say something about the effect of speed on muscle stress should be discussed.

We have now provided additional details, (see responses above) to these concerns. For instance, we added a supplementary figure showing the speed distribution per individual. The primary reviewer concern (that each kangaroo travelled at a single speed) was due to a miscommunication around the terminology “preferred” which has now been corrected.

We now elaborate in the results why we are not very concerned that EMA is insignificant. The statistical insignificance of EMA is ultimately due to the insignificance of the direct measurement of R , however, we now better explain in the results why we believe that this statistical insignificance is due to error/noise of the measurement which is relatively large compared to the effect size. Indirect indications of how R may increase with speed (via ankle height from the ground) are statistically significant. Lines 177-185.

We consider this worth reporting because, for instance, an 18% change in EMA will be undetectable by measurement, but corresponds to an 18% change in tendon stress which is measurable and physiologically significant (safety factor would decrease from 2 to 1.67). We presented both significant and insignificant results for transparency.

We have also discussed this within a revised limitations section of the manuscript (Line 311328).

Reviewer #1 (Recommendations For The Authors):

Title: I would cut the first half of the title. At least hedge it a bit. "Clues" instead of "Unlocking the secrets".

We have revised the title to: “Postural adaptations may contribute to the unique locomotor energetics seen in hopping kangaroos”

In my comments, ... typically indicates a stylistic change suggested to the text.

Overall, the paper covers speed and size. Unfortunately, the authors were not 100% consistent in the order of presenting size then speed, or speed then size. Just choose one and stick with it.

We have attempted to keep the order of presenting size and speed consistent, however there are several cases where this would reduce the readability of the manuscript and so in some cases this may vary.

One must admit that there is a lot of vertical scatter in almost all of the plots. I understand that these animals were not in a lab on a treadmill at a controlled speed and the animals wear fur coats so marker placements vary/move etc. But the spread is quite striking, e.g. Figure 5a the span at one speed is almost 10x. Can the authors address this somewhere? Limitations section?

The variation seen likely results from attempting to display data in a 2D format, when it is in fact the result of multiple variables, including speed, mass, stride frequency and subject specific lengths. Slight variations in these would be expected to produce some noise around

the mean, and I think it's important to consider this while showing the more dominant effects.

In many locations in the manuscript, the term "work" is used, but rarely if ever specified that this is the work "per hop". The big question revolves around the rate of metabolic energy consumption (i.e. energy per time or average metabolic power), one must not forget that hop frequency changes somewhat across speed, so work per hop is not the final calculation.

Thank you for this comment. We have now explicitly stated work per hop in figure captions and in the results (line 208). The change in stride frequency at this range of speeds is very small, particularly compared to the variance in stride frequency (Suppl. Fig. 1d), which is consistent with other researchers who found that stride frequency was constant or near constant in macropods at analogous speeds (e.g. Dawson and Taylor 1973, Baudinette et al. 1987).

Line 61is likely related.

Added "likely" (line 59)

Line 86 I think the Allen reference is incomplete. Wasn't it in J Exp Biology?

Thank you. Changed.

Line 122 ... at faster speeds and in larger individuals.

Changed: "We hypothesised that (i) the hindlimb would be more crouched at faster speeds, primarily due to the distal hindlimb joints (ankle and metatarsophalangeal), independent of changes with body mass" (Line 121-122).

Line 124 I found this confusing. Try to re-word so that you explain you mean more work done by the tendons and less by the ankle musculature.

Amended: "changes in moment arms resulting from the change in posture would contribute to the increase in tendon stress with speed, and may thereby contribute to energetic savings by increasing the amount of positive and negative work done by the ankle without requiring additional muscle work" (Line 123)

Line 129 hopefully "braking" not "breaking"!

Thank you. Fixed. (Line 130)

Line 129 specify fore-aft horizontal force.

Added "fore-aft" to "negative fore-aft horizontal component" (Line 130-131)

Line 130 add something like "of course" or "naturally" since if there is zero fore-aft force, the GRF vector of course must be vertical.

Added "naturally" (Line 132)

Line 138 clarify that this section is all stance phase. I don't recall reading any swing phase data.

Changed to: "Kangaroo hindlimb stance phase kinematics varied..." (Line 141)

Line 143 and elsewhere. I found the use of dorsiflexion and plantarflexion confusing. In Figure 3, I see the ankle never flexing more than 90 degrees. So, the ankle joint is always in something of a flexed position, though of course it flexes and extends during contact. I urge the authors to simplify to flexion/extension and drop the plantar/dorsi.

We have edited this section to describe both movements as greater extension (plantarflexion). (Line 147). We have further clarified this in the figure caption for figure 3.

Line 147 ...changes were...

Fixed, line 150

Line 155 I'm a bit confused here. Are the authors calculating some sort of overall EMA or are they saying all of the individual joint EMAs all decreased?

Thank you, we clarified that it is at the ankle. Line 158

Line 158 since kangaroos hop and are thus positioned high and low throughout the stance phase, try to avoid using "high" and "low" for describing variables, e.g. GRF or other variables. Just use "greater/greatest" etc.

Thanks for this suggestion. We have changed "higher" into "greater" where appropriate throughout the manuscript e.g. line 161

Lines 162 and 168 same comment here about "r" and "R". Do you mean ankle or all joints?

Clarified that it is the gastrocnemius and plantaris r, and the R to the ankle. (Lines 164-165)

Line 173 really, ankle height?

Added: ankle height is "vertical distance from the ground". Line 177

Line 177 is this just the ankle r?

Added "of the ankle" line 158 and "Achilles" line 187

Line 183 same idea, which tendon/tendons are you talking about here?

Added "Achilles" to be more clear (Line 187)

Line 195 substitute "converted" for "transferred".

Done (Line 210)

Line 223 why so vague? i.e. why use "may"? Believe in your data. ...stress was also modulated by changes....

Changed "may" to "is"

Line 229 smaller ankle EMA (especially since you earlier talked about ankle "height").

Changed “lower” to “smaller” Line 254

| *Line 2236 ...and return elastic energy...*

Added "elastic" line 262

| *Line 244 IMPORTANT: Need to explain this better! I think you are saying that the net work at the ankle is staying the same across speed, BUT it is the tendons that are storing and returning that work, it's not that the muscles are doing a lot of negative/positive work.*

Changed: “The consistent net work observed among all speeds suggests the ankle extensor muscle-tendon units are performing similar amounts of ankle work independent of speed, which would predominantly be done by the tendon.” Line 270-272)

| *Line 258-261 I think here is where you are over-selling the data/story. Although you do say "a" mechanism (and not "the" mechanism, you still need to deal with the cost of generating more force and generating that force faster.*

We removed this sentence and replaced it with a discussion of the cost of generating force hypothesis, and alternative scenarios for the how force and metabolics could be uncoupled.

| *Line 278 "the" tendon? Which tendon?*

Added "Achilles"

| *Line 289. I don't think one can project into the past.*

Changed “projected” to “estimated”

| *Line 303 no problem, but I've never seen a paper in biology where the authors admit they don't know what species they were studying!*

Can't be helped unfortunately. It is an old dataset and there aren't photos of every kangaroo. Fortunately, from the grey and red kangaroos we can distinguish between, we know there are no discernible species effects on the data.

| *Lines 304-306 I'm not clear here. Did you use vertical impulse (and aerial time) to calculate body weight? Or did you somehow use the braking/propulsive impulse to calculate mass? I would have just put some apples on the force plate and waited for them to stop for a snack.*

Stationary weights were recorded for some kangaroos which did stand on the force plate long enough, but unfortunately not all of them were willing to do so. In those cases, yes, we used impulse from steady-speed trials to estimate mass. We cross-checked by estimated mass from segment lengths (as size and mass are correlated). This is outlined in the first paragraph of the methods.

| *Lines 367 & 401 When you use the word "scaled" do you mean you assumed geometric similarity?*

No, rather than geometric scaling, we allowed scaling to individual dimensions by using the markers at midstance for measurements. We have amended the paragraph to clarify that the

shape of the kangaroo changes and that mass distribution was preserved during the shape change (line 441-446)

| *Lines 381-82 specify "joint work"*

Added "joint work" (Line 457)

| *Figure 1 is gorgeous. Why not add the CF equation to the left panel of the caption?*

We decided to keep the information in the figure caption. "Total leg length was calculated as the sum of the segment lengths (solid black lines) in the hindlimb and compared to the pelvito-toe distance (dashed line) to calculate the crouch factor"

| *Figure 2 specify Horizontal fore-aft.*

Done

| *Figure 3g I'd prefer the same Min. Max Flexion vertical axis labels as you use for hip & knee.*

While we appreciate the reviewer trying to increase the clarity of this figure, we have left it as plantar/dorsi flexion since these are recognised biomechanical terms. To avoid confusion, we have further defined these in the figure caption "For (f-g), increased plantarflexion represents a decrease in joint flexion, while increased dorsiflexion represents increased flexion of the joint."

| *Figure 4. I like it and I think that you scaled all panels the same, i.e. 400 W is represented by the same vertical distance in all panels. But if that's true, please state so in the Caption. It's remarkable how little work occurs at the hip and knee despite the relatively huge muscles there.*

Is it true that the y axes are all at the same scale. We have added this to the caption.

| *Figure 5 Caption should specify "work per hop".*

Added

| *Figure 7 is another beauty.*

Thank you!

| *Supplementary Figure 3 is this all ANKLE? Please specify.*

Clarified that it is the gastrocnemius and plantaris r, and the R to the ankle.

Reviewer #2 (Recommendations For The Authors):

To 'unlock the secrets of kangaroo locomotor energetics' I expected the authors to measure the secretive outcome variable, metabolic rate using laboratory measures. Rather, the authors relied on reviewing historic metabolic data and collecting biomechanics data across different animals, which limits the conclusions of this manuscript.

We have revised to the title to make it clearer that we are investigating a subset of the energetics problem, specifically posture. “Postural adaptations may contribute to the unique locomotor energetics seen in hopping kangaroos.” We have also substantially modified the discussion to temper the conclusions from the paper.

After reading the hypothesis, why do the authors hypothesize about joint flexion and not EMA? Because the following hypothesis discusses the implications of moment arms on tendon stress, EMA predictions are more relevant (and much more discussed throughout the manuscript).

Ankle and MTP angles are the primary drivers of changes in r , R & thus, EMA. We used a two part hypothesis to capture this. We have rephased the hypotheses: “We hypothesised that (i) the hindlimb would be more crouched at faster speeds, primarily due to the distal hindlimb joints (ankle and metatarsophalangeal), independent of changes with body mass, and (ii) changes in moment arms resulting from the change in posture would contribute to the increase in tendon stress with speed, and may thereby contribute to energetic savings by increasing the amount of positive and negative work done by the ankle without requiring additional muscle work.”

If there were no detectable effects of speed on EMA, are kangaroos mechanically like other animals (Biewener Science 89 & JAP 04) who don't vary EMA across speeds? Despite no detectable effects, the authors state [lines 228-229] "we found larger and faster kangaroos were more crouched, leading to lower ankle EMA". Can the authors explain this inconsistency? Lines 236 "Kangaroos appear to use changes in posture and EMA". I interpret the paper as EMA does not change across speed.

Apologies, we did not sufficiently explain this originally. We now explain in the results our reasoning behind our belief that EMA and R may change with speed. “If we consider the nonsignificant relationship between R (and EMA) and speed to indicate that there is no change in R , then it conflicts with the ankle height and CoP result. Taking both into account, we think it is more likely that there is a small, but important, change in R , rather than no change in R with speed. It may be undetectable because we expect small effect sizes compared to the measurement range and measurement error (Suppl. Fig. 3h), or be obscured by a similar change in R with body mass. R is highly dependent on the length of the metatarsal segment, which is longer in larger kangaroos (1 kg BM corresponded to ~1% longer segment, $P < 0.001$, $R^2 = 0.449$). If R does indeed increase with speed, both R and r will tend to decrease EMA at faster speeds.” (Line 177-185)

Lines 335-339: "We assumed the force was applied along phalanx IV and that there was no medial or lateral movement of the centre of pressure (CoP)". I'm confused, did the authors not measure CoP location with respect to the kangaroo limb? If not, this simple estimation undermines primary results (EMA analyses).

We have changed “The anterior or posterior movement of the CoP was recorded by the force plate” to read: “The fore-aft movement of the CoP was recorded by the force plate within the motion capture coordinate system” (Line 406-407) and added more justification for fixing the CoP movement in the other axis: “It was necessary to assume the CoP was fixed in the mediolateral axis because when two feet land on the force plate, the lateral forces on each foot are not recorded, and indeed cancel if the forces are symmetrical (i.e. if the kangaroo is hopping in a straight path and one foot is not in front of the other). We only used symmetrical trials to ensure reliable measures of the anterior-posterior movement of the CoP.” (Line 408-413)

The introduction makes many assertions about the generalities of locomotion and the relationship between mechanics and energetics. I'm afraid that the authors are selectively choosing references without thoroughly evaluating alternative theories. For example, Taylor, Kram, & others have multiple papers suggesting that decreasing EMA and increasing muscle force (and active muscle volume) increase metabolic costs during terrestrial locomotion. Rather, the authors suggest that decreasing EMA and increasingly high muscle force at faster speeds don't affect energetics unless muscle work increases substantially (paragraph 2)? If I am following correctly, does this theory conflict with active muscle volume ideas that are peppered throughout this manuscript?

Yes, as you point out, the same mechanism does lead to different results in kangaroos vs humans, for instance, but this is not a contradiction. In all species, decreasing EMA will result in an increase in muscle force due to less efficient leverage (i.e. lower EMA) of the muscles, and the muscle-tendon unit will be required to produce more force to balance the joint moment. As a consequence, human muscles activate a greater volume in order for the muscle-tendon unit to increase muscle work and produce enough force. We are proposing that in kangaroos, the increase in work is done by the achilles tendon rather than the muscles. Previous research suggests that macropod ankle muscles contract isometrically or that the fibres do not shorten more at faster speeds i.e. muscle work does not increase with speed. Instead, the additional force seems to come from the tendon storing and subsequently returning more strain energy (indicated by higher stress). We found that the increase in tendon stress comes from higher ground force at faster speeds, and from it adopting a more crouched posture which increases the tendons' stresses compared to an upright posture for a given speed (think of this as increasing the tendon's stress capacity). We have substantially revised the discussion to highlight this.

Similarly, does increased gross or net tendon mechanical energy storage & return improve hopping energetics? Would more tendon stress and strain energy storage with a given hysteresis value also dissipate more mechanical energy, requiring leg muscles to produce more net work? Does net or gross muscle work drive metabolic energy consumption?

Based on the cost of generating force hypothesis, we think that gross muscle work would be linked to driving metabolic energy consumption. Our idea here is that the total body work is a product of the work done by the tendon and the muscle combined. If the tendon has the potential to do more work, then the total work can increase without muscle work needing to increase.

The results interpret speed effects on biomechanics, but each kangaroo was only collected at 1 speed. Are inter-animal comparisons enough to satisfy this investigation?

We have added a figure (Suppl Fig 9) to demonstrate the distribution of speed and number of trials per kangaroo. We have also removed "preferred" from the manuscript as this seems to cause confusion. Most kangaroos travelled at a range of "casual" speeds.

Abstract: Can the authors more fully connect the concept of tendon stress and low metabolic rates during hopping across speeds? Surely, tendon mechanics don't directly drive the metabolic cost of hopping, but they affect muscle mechanics to affect energetics.

Amended to: " This phenomenon may be related to greater elastic energy savings due to increasing tendon stress; however, the mechanisms which enable the rise in stress, without additional muscle work remain poorly understood." (Lines 25-27).

The topic sentence in lines 61-63 may be misleading. The ensuing paragraph does not substantiate the topic sentence stating that ankle MTUs decouple speeds and energetics.

We added "likely" to soften the statement. (Line 59)

Lines 84-86: In humans, does more limb flexion and worse EMA necessitate greater active muscle volume? What about muscle contractile dynamics - See recent papers by Sawicki & colleagues that include Hill-type muscle mechanics in active muscle volume estimates.

Added: "Smaller EMA requires greater muscle force to produce a given force on the ground, thereby demanding a greater volume of active muscle, and presumably greater metabolic rates than larger EMA for the same physiology". (Line 80-82)

Lines 106: can you give the context of what normal tendon safety factors are?

Good idea. Added: "far lower than the typical safety factor of four to eight for mammalian tendons (Ker et al. 1988)." Line 106-107

I thought EMA was relatively stable across speeds as per Biewener [Science & JAP '04]. However the authors gave an example of an elephant to suggest that it is typically inversely related to speed. Can the authors please explain the disconnect and the most appropriate explanation in this paragraph?

Knee EMA in particular changed with speed in Biewener 2004. What is "typical" probably depends on the group of animals studied; e.g., cursorial quadrupedal mammals generally seem to maintain constant EMA, but other groups do not.

These cases are presented to show a range of consequences for changing EMA (usually with mass, but sometimes with speed). We have made several adjustments to the paragraph to make this clearer. Lines 85-93.

The results depend on the modeled internal moment arm (r). How confident are the authors in their little r prediction? Considering complications of joint mechanics in vivo including muscle bulging. Holzer et al. '20 Sci Rep demonstrated that different models of the human Achilles tendon moment arm predict vastly different relationships between the moment arm and joint angle.

Our values for r and EMA closely align with previous papers which measured/calculate these values in kangaroos, such as Kram 1998, and thus we are confident in our interpretation.

This is a misleading results sentence: Small decreases in EMA correspond to a nontrivial increase in tendon stress, for instance, reducing EMA from 0.242 (mean minimum EMA of the slow group) to 0.206 (mean minimum EMA of the fast group) was associated with an ~18% increase in tendon stress. The authors could alternatively say that a ~15% decrease in EMA was associated with an ~18% increase in tendon stress, which seems pretty comparable.

Thank you for pointing this out, it is important that it is made clearer. Although the change in relative magnitude is approximately the same (as it should be), this does not detract from the importance. The "small decrease in EMA" is referring to the absolute values, particularly in respect to the measurement error/noise. The difference is small enough to have been undetectable with other methods used in previous studies. We have amended the sentence to clarify this.

It now reads: “Subtle decreases in EMA which may have been undetected in previous studies correspond to discernible increases in tendon stress. For instance, reducing EMA from 0.242 (mean minimum EMA of the slow group) to 0.206 (mean minimum EMA of the fast group) was associated with an increase in tendon stress from ~50 MPa to ~60 MPa, decreasing safety factor from 2 to 1.67 (where 1 indicates failure), which is both measurable and physiologically significant.” (Line 195-200)

Lines 243-245: "The consistent net work observed among all speeds suggests the ankle extensors are performing similar amounts of ankle work independent of speed." If this is true, and presumably there is greater limb work performed on the center of mass at faster speeds (Donelan, Kram, Kuo), do more proximal leg joints increase work and energy consumption at faster speeds?

The skin over the proximal leg joints (knee and hip) moves too much to get reliable measures of EMA from the ratio of moment arms. This will be pursued in future work when all muscles are incorporated in the model so knee and hip EMA can be determined from muscle force.

We have added limitations and considerations paragraph to the manuscript: “Finally, we did not determine whether the EMA of proximal hindlimb joints (which are more difficult to track via surface motion capture markers) remained constant with speed. Although the hip and knee contribute substantially less work than the ankle joint (Fig. 4), the majority of kangaroo skeletal muscle is located around these proximal joints. A change in EMA at the hip or knee could influence a larger muscle mass than at the ankle, potentially counteracting or enhancing energy savings in the ankle extensor muscle-tendon units. Further research is needed to understand how posture and muscles throughout the whole body contribute to kangaroo energetics.” (Line 321-328)

Lines 245-246: "Previous studies using sonomicrometry have shown that the muscles of tammar wallabies do not shorten considerably during hops, but rather act near-isometrically as a strut" Which muscles? All muscles? Extensors at a single joint?

Added "gastrocnemius and plantaris" Line 164-165

Lines 249-254: "The cost of generating force hypothesis suggests that faster movement speeds require greater rates of muscle force development, and in turn greater cross-bridge cycling rates, driving up metabolic costs (Taylor et al. 1980, Kram and Taylor 1990). The ability for the ankle extensor muscle fibres to remain isometric and produce similar amounts of work at all speeds may help explain why hopping macropods do not follow the energetic trends observed in quadrupedal species." These sentences confuse me. Kram & Taylor's cost of force-generating hypothesis assumes that producing the same average force over shorter contact times increases metabolic rate. How does 'similar muscle work' across all speeds explain the ability of macropods to use unique energetic trends in the cost of force-generating hypothesis context?

Thank you for highlighting this confusion. We have substantially revised the discussion clarify where the mechanisms presented deviate from the cost of generating force hypothesis. Lines 270-309

Reviewer #3 (Recommendations For The Authors):

In addition to the points described in the public review, I have additional, related, specific comments:

(1) Results: Please refer to the hypotheses in the results, and relate the the findings back to the hypotheses.

We now relate the findings back to the hypotheses

Line 142 “In partial support of hypothesis (i), greater masses and faster speeds were associated with more crouched hindlimb postures (Fig. 3a,c).”

Lines 205-206: “The increase in tendon stress with speed, facilitated in part by the change in moment arms by the shift in posture, may explain changes in ankle work (c.f. Hypothesis (ii)).”

(2) Results: please provide the main statistical results either in-line or in a table in the main text.

We (the co-authors) have discussed this at length, and have agreed that the manuscript is far more readable in the format whereby most statistics lie within the supplementary tables, otherwise a reader is met with a wall of statistics. We only include values in the main text when the magnitude is relevant to the arguments presented in the results and discussion.

(3) Line 140: Describe how 'crouched' was defined.

We have now added a brief definition of ‘Crouch factor’ after the figure caption. (Line 143) (Fig. 3a,c; where crouch factor is the ratio of total limb length to pelvis to toe distance).

(4) Line 162: This seems to be a main finding and should be a figure in the main text not supplemental. Additionally, Supplementary Figures 3a and b do not show this finding convincingly There should be a figure plotting r vs speed and r vs mass.

The combination of r and R are represented in the EMA plot in the main text. The r and R plots are relegated to the supplementary because the main text is already very crowded. Thank you for the suggestion for the figure plotting r and R versus speed, this is now included as Suppl. Fig. 3h

(5) Line 166: Supplementary Figure 3g does not show the range of dorsiflexion angles as a function of speed. It shows r vs dorsiflexion angle. Please correct.

Thanks for noticing this, it was supposed to reference Fig 3g rather than Suppl Fig 3g in the sentence regarding speed. We have fixed this, Line 170.

We had added a reference to Suppl Fig 3 on Line 169 as this shows where the peak in r with ankle angle occurs (114.4 degrees).

(6) Line 184: Where are the statistical results for this statement?

The relationship between stress and EMA does not appear to be linear, thus we only present R^2 for the power relationship rather than a p-value.

(7) Line 192: The authors should explain how joint work and power relate/support the overall hypotheses. This section also refers to Figures 4 and 5 even though Figures 6 and 7 have already been described. Please reorganize.

We have added a sentence at the end of the work and power section to mention hypothesis (ii) and lead into the discussion where it is elaborated upon.

“The increase in positive and negative ankle work may be due to the increase in tendon stress rather than additional muscle work.” Line 219-220 We have rearranged the figure order.

(8) The statistics are not reported in the main text, but in the supplementary tables. If a result is reported in the main text, please report either in-line or with a table in the main text.

We leave most statistics in the supplementary tables to preserve the readability of the manuscript. We only include values in the main text when the magnitude is relevant to the arguments raised in the results and discussion.

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